

Five/Ten-Slot Module Service

LC13-600

LCN Service - 1

Five/Ten-Slot Module Service

**LC13-600
Release 684
CE Compliant
March 2013**

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About This Publication

This publication provides instructions for system service personnel to service any Five/Ten-Slot Module. It will help you to determine how to perform service that is required for Five/Ten-Slot Modules, identify spare parts, and provide disassembly/assembly instructions for replacing the required part.

The publication supports the *Universal Station Service*, *Universal Station^X Service*, and *History Module Service* manuals.

The publication supports TPS Network software release 530 and later, and CE Compliant hardware.

Any equipment designated as "CE Compliant" complies with the European Union EMC and Health and Safety Directives. All equipment shipping into European Union countries after January 1, 1996 requires this type of compliance—denoted by the "CE Mark."

ATTENTION

With TPN R684, Enhanced Universal Control Network (EUCN) is introduced. In EUCN, the traditional coaxial cable system is no longer utilized and is replaced with a Fault Tolerant Ethernet (FTE) based communication network.

Throughout the document, unless specifically mentioned, the terms HPM, NIM, and UCN are interchangeably used with the terms EHPM, ENIM, and EUCN respectively.

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Standard Symbols

Scope

The following defines standard symbols used in this publication.

ATTENTION

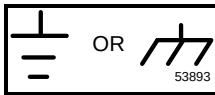
Notes inform the reader about information that is required, but not immediately evident.

CAUTION

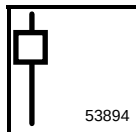
Cautions tell the user that damage may occur to equipment if proper care is not exercised.

WARNING

Warnings tell the reader that potential personal harm or serious economic loss may happen if instructions are not followed.



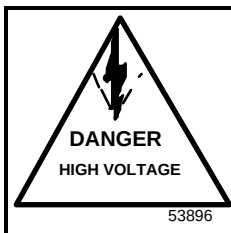
Ground connection to building safety ground



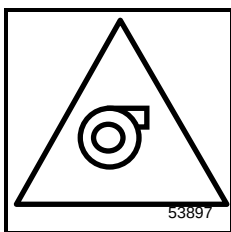
Ground stake for building safety ground



Electrical Shock Hazard—can be lethal



Electrical Shock Hazard—can be lethal



Rotating Fan—can cause personal injury

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INTRODUCTION

Section 1

1.1 OVERVIEW

This manual provides detailed instructions for maintenance, testing, troubleshooting and repair of the Five- and Ten-Slot Modules (part of the TPS system). A typical Five-Slot Module is shown in Figure 1-1. The troubleshooting, disassembly, and assembly procedures are effective down to the optimum replaceable-unit (ORU) level. A spare parts list (subsection 5.1) contains ORU parts and is keyed to a module exploded view that is also used with the disassembly and assembly procedures.

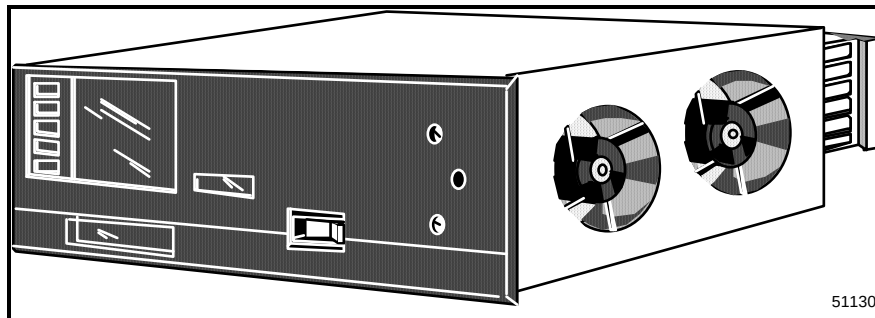


Figure 1-1 — Five-Slot Module

1.2 RELATED PUBLICATIONS

The following publications apply to the TPS system and should be referred to as required.

Title	Binder
LCN System Installation	LCN Installation
LCN System Checkout	LCN Installation
Universal Station Service	LCN Service - 1
Universal Station (Ergonomic) Service	LCN Service - 1
Universal Station ^X Service	Universal Station ^X
Universal Station ^X (Ergonomic) Service	Universal Station ^X
History Module Service	LCN Service - 2
Dual Node Module Service	LCN Service - 2
Maintenance Test Operations	LCN Service - 1
System Maintenance Guide	LCN Service - 1
Application Module ^X Service	Application Module ^X

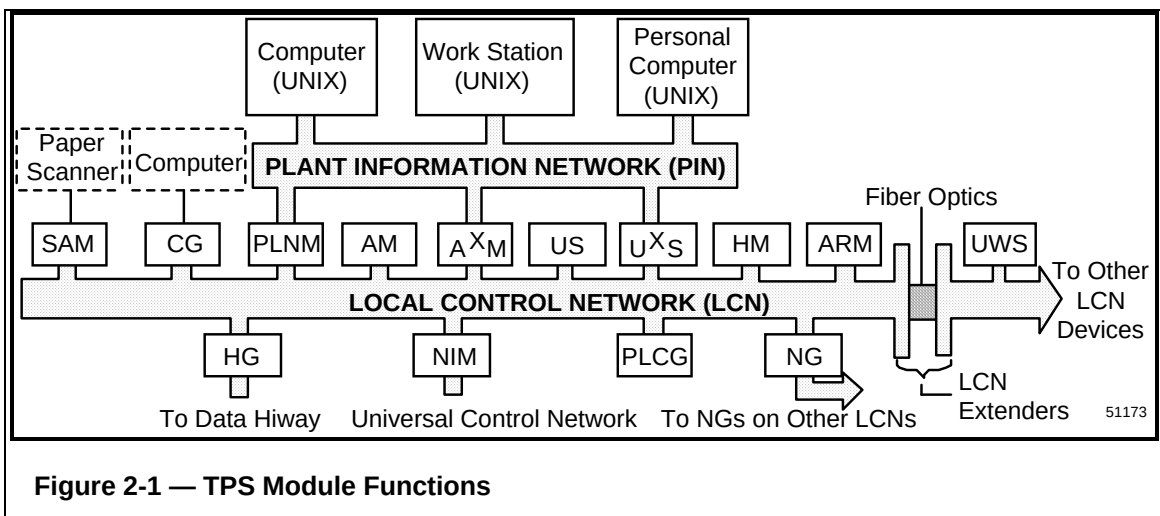
MODULE DESCRIPTION

Section 2

2.1 GENERAL DESCRIPTION

The Five-Slot and Ten-Slot Modules are configured to perform various functions, such as Applications, Gateways, History, and Universal Stations in the TPS system. Each module occupies a specific address (node) on the Local Control Network (LCN). Figure 2-1 illustrates the module functions within a typical TPS system.

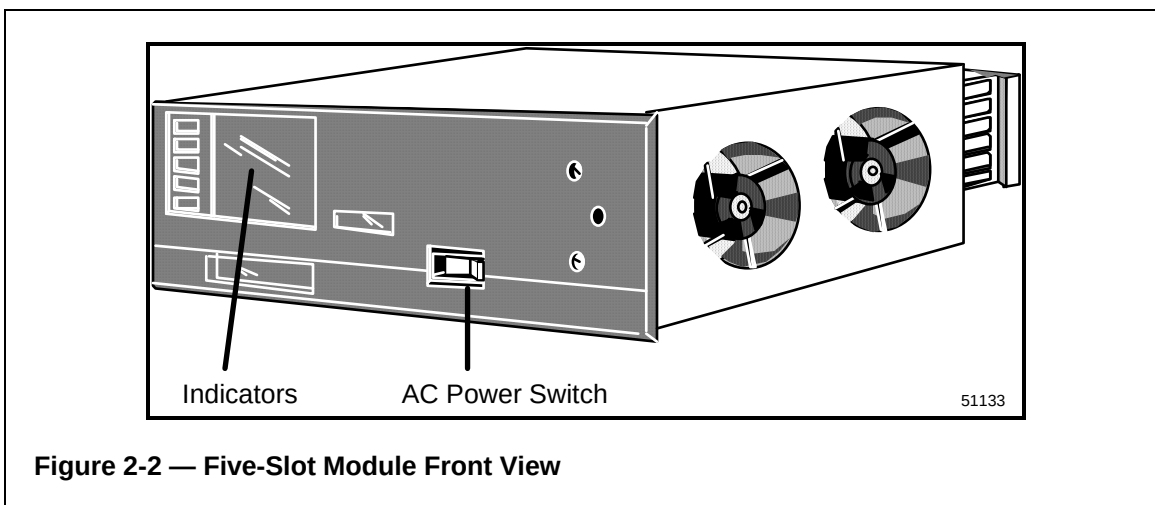
The configuration flexibility is because of the inter-changeability of the component card assemblies (boards) in the module. In addition to the boards within the module, there are also companion input-output (I/O) boards mounted in a chassis on the rear of the module. The I/O boards facilitate interconnection of the module to the LCN or with the Data Hiway, Universal Control Network (UCN), external devices, and communication lines. The module also contains a power supply, located in the bottom of the unit, and a cooling-fan assembly, located on the right side.



The modules are designed to be mounted in the Honeywell TPS system equipment cabinet and Universal Station console bays that conform to the industry standard 19-inch rack mount. If mounting racks other than Honeywell's are used, care must be taken to ensure that the cooling-fan air flow is not blocked and that adequate air filtration is provided.

2.1.1 Controls and Features

The front of the Five/Ten-Slot Module has a single ON/OFF ac power switch that controls the alternating current to the power supply. This switch is shown in Figure 2-2. When the power switch is in the ON position, LED indicators can be seen through the see-through portion of the front cover panel.



The rear side of the module has several connectors that are used as shown in Table 2-1.

Table 2-1 — Five-Slot Module Connector Usage

CONNECTOR	USAGE
J2	Ac Power Cord
J5	Remote Reset
J7	Precision Clock Input
J8	+5 Vdc and +12 Vdc Power for Peripherals
J9	+5 Vdc and +12 Vdc Power for Peripherals
J10	24 Vdc for Fan Power on Cartridge/Floppy/History Module Assemblies

When a Five-Slot Module that is located in a Universal Station (US) that has a floppy drive or cartridge drive, the power supply (51201436) can be used as a power source to drive the floppy drive or cartridge drive and to power the cooling fan for the cartridge drive. Connector J8 is used for the drive power, but J9 provides the same power connection.

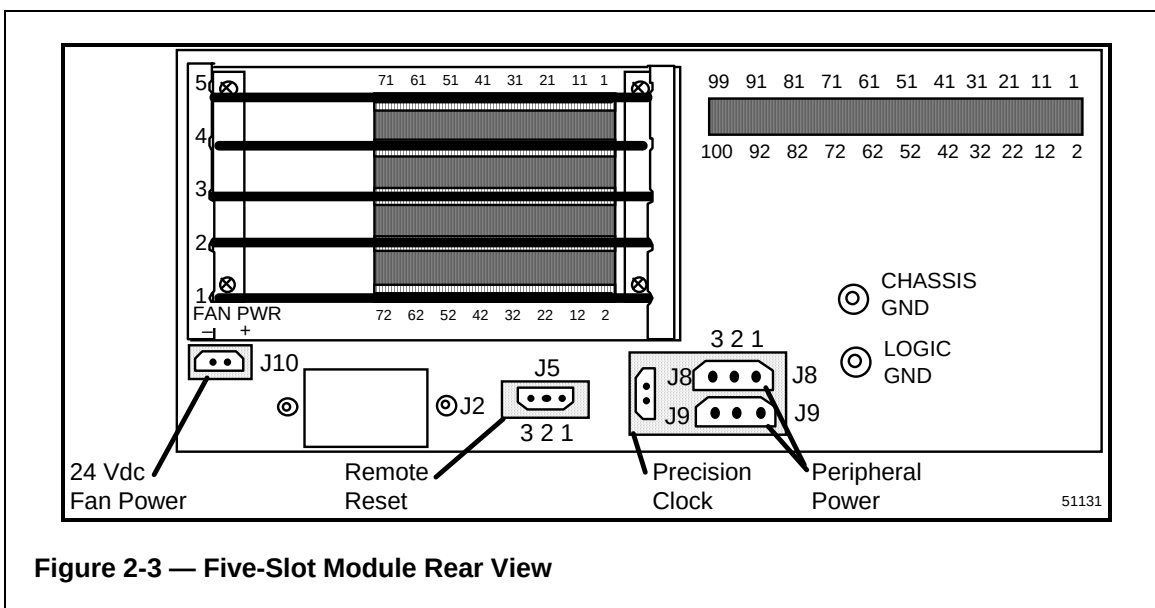


Figure 2-3 — Five-Slot Module Rear View

The CE compliant Five-Slot Module is capable of holding both I/O board designs.

- The original design I/O boards function as they always have.
- The CE compliant I/O boards mount to the card file with screws. This grounds the I/O board face plate and the cable shields that connect to the face plate.

The CE compliant module is the same in all other respects.

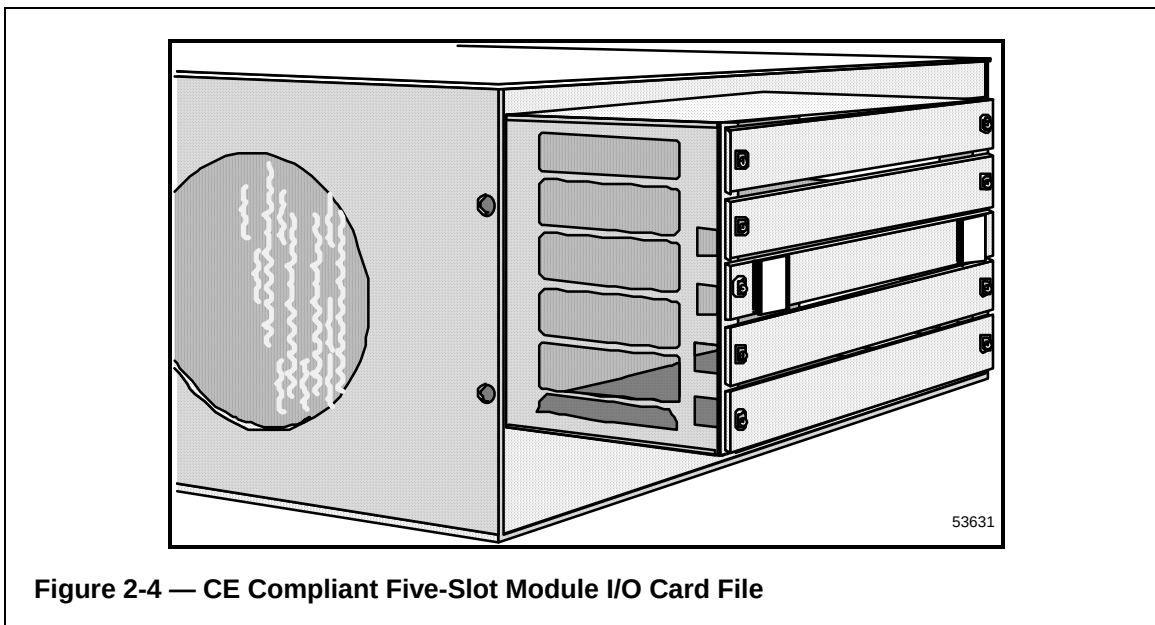


Figure 2-4 — CE Compliant Five-Slot Module I/O Card File

2.2 MODULE CONFIGURATIONS

Module functions are determined by the board complement assigned to them. The board slots on both the front and rear of the unit are numbered from 1-to-5 or 1-to-10 with slot number one at the bottom. Tables 2-7 through 2-29 list the board configurations for the factory-built modules and upgraded modules that will operate with the latest R500 software. Please note that some types of modules can be installed in either Five- or Ten-Slot Module chassis.

Table 2-2 — Configurations and Peripherals

Configuration/ Peripheral	Description	Boards Used
Standard Performance	68000 Processor	EMPU
High Performance	68020 Processor	HMPU, HPK2-x, or K2LCN
Very High Performance	68040 Processor	K4LCN
Interlaced Monitor	“Standard” 19 inch color monitor mounted in the Universal Station and driven by separate red, green, and blue signals on coaxial cables.	PIC, VDG, & VDG I/O or PDG & PDG I/O* or EPDG & PDG I/O**
13” Monitor	A stand-alone color monitor, smaller but electrically similar to the interlaced monitor. Driven by three TTL signals. Used in the Universal Work Station (UWS).	PDG & PDG I/O ^{TTL*}
Noninterlaced Monitor	An “Enhanced Display” 19 inch color monitor mounted in the Universal Station offering a sharper, clearer picture than the interlaced monitor. It is driven by a ribbon cable.	EPDG & EPDG I/O or EPDGP I/O
14” Monitor	A stand-alone color monitor, smaller but electrically similar to the noninterlaced monitor. Used in the Enhanced UWS.	EPDG & EPDG I/O or EPDGP I/O
Lower Tier	A part of the Universal Station cabinet containing a monitor. It is mounted in front of the operator, just above the keyboard.	
Upper Tier	An additional location on the Universal Station for a monitor to be mounted above the lower tier. No keyboard is available. A separate Five-10 slot module is required.	
21” FST Monitor	Monitor for Universal Station ^X that is a flat square tube with high resolution graphics (1280 x 1024 pixels).	TPDG TPDGX TPDG I/O
Hard Disk Drive Tray (HDDT)	Holds one or two disk drives (525 MB, 1 GB, or 2 GB) for the WSI2 coprocessor.	WSI WSI2 I/O

* There are 2 different types of PDG I/O boards—see Table 2-5.

** If an EPDG board drives a PDG I/O board, it may drive an interlaced monitor.

Module configurations change for several reasons, some of which are:

- Hardware components (such as processors, hard disk drives, cartridge drives, and others) are introduced which offer faster operation, more storage, or more functionality.
- The basic design of peripherals (such as monitors, drives, keyboards, and others) may change, requiring modifications in the electrical circuits servicing them.
- A new software release containing greater functionality may require more execution speed or more memory.

Table 2-2 describes hardware configurations and peripherals which you may encounter in this manual.

2.2.1 Board Application Notes

The board types listed in Tables 2-2 through 2-5 are the current production board types. Table 2-3 briefly describes the features of the newest boards and the minimum software releases that they require.

Many boards, in addition to those listed in Table 2-3, still operate satisfactorily with R4xx software. All of these boards, including brief descriptions and part numbers, are listed in Tables 2-4 and 2-5.

Table 2-4 lists the functional boards that are installed at the front of the 5/10 slot module. Table 2-5 lists the I/O and special-purpose paddle boards that are installed in the back of the module. An I/O board is normally installed directly behind the functional board it serves, as shown in the configuration tables in subsections 2.2.2 through 2.2.11. Special-purpose boards can generally be installed in any unused I/O slot, but check the appropriate service manual if you don't know where a board will operate satisfactorily.

NOTE

Under certain circumstances, the K2LCN OR K4LCN processor board can be used to replace HPK2-x and EMPU processors. In the case of EMPU replacement, there may be a performance enhancement. The restrictions and procedures involved in this replacement are covered in detail in Appendix B.

Table 2-3 — Board Replacement Notes

Board Type	Description	Minimum Software Release
EMEM	1 MB memory board used as main memory with MCPU and EMPU boards. This board also is used as additional memory with HMPU and HPK2-x boards.	200
EPDG	With the PDG I/O board, operates as a PDG with the interlaced monitor, and can replace the PDG or VDG/PIC.	200
EPDG	With the EPDG or EPDGP I/O board, operates as an EPDG with the noninterlaced monitor. Also has a Cartridge Disk interface.	210M1
EPDGP I/O	EPDGP I/O board (Tab-300) with jumper (J11) for DIRECT or INVERT of vertical sync signal. Vertical sync for 21" FST monitors in XC3000 consoles require inverted sync signal.	300
HMPU	68020 processor and memory (2 MB) board replaced by HPK2-x in current production (except in redundant AMs). The HMPU board also contains a floating-point coprocessor.	200 (HG) 210 (All)
HPK2-2	68020 processor with 2 MB of on-board memory that replaces an HMPU, EMPU, and MCPU in all modules except redundant AMs (MCPU is not recommended for R4xx).	210M1
HPK2-3	68020 processor and memory (same as HPK2-2 with 3 MB).	230
K2LCN-x	68020 combined processor, memory and LCN interface (eliminates LCN board and external memory boards). Available with x = 2, 4, or 8 MB of on board memory.	320
K4LCN-x	68040 combined processor, memory and LCN interface (eliminates LCN board and external memory boards). Available with x = 4, 8, or 16 MB of on board memory, ordered separately.	500
LLCN	Replacement for LCN board. Uses low power consumption circuits. Operates with LCN I/O board.	Any
QMEM-1	This board type has been withdrawn. Use EMEM for additional memory of 1 megaword increments.	200
QMEM-2, 3, 4	Quad Memory board of 2, 3 or 4 MB capacity. Memory capacity cannot be increased in the field. Each has a unique ID.	300
TPDG	With TPDG I/O drives a 21" FST monitor in a Universal Station ^X .	410/U ^X S R100
TPDGX	With TPDG I/O drives a 21" FST monitor in a Universal Station ^X requires WSI2.	410/U ^X S R200
WSI	Release 100/110 Work Station Interface for Universal Station ^X .	410/U ^X S R100
WSI2	Release 200 Work Station Interface for Universal Station ^X . Board has replaceable memory daughter boards in sizes = 2-16 MB, 3 2 MB, 2-32 MB, 64 MB, or 2-64 MB.	410/U ^X S R200
WSI2R I/O	Replaces WSI I/O in R100 units updated to R200 with WSI2 board	R200
WSI2 I/O	Used with WSI2 board in new build units and units upgraded from US to UXS (R200) by upgrade kits MP-ZUXCC2 or MP-ZUXNC2.	R200

Table 2-4 — Functional Board Types

Board Type	Description	Part Number
AMR	Redundant AM Interface	51401070-100
CLI	Communications Line Interface	80360206-001
CNI	Communications Network Interface	51401088-100
DHI	Data Hiway Interface	51400700-100
EAMR	Enhanced Application Module Redundancy	51401996-100
EMEM	1 MB Enhanced Memory	51400910-100
EMPU	Enhanced Microprocessor board	51400901-100
EPNI	Enhanced Network Interface	51401583-100
EPNI2	Enhanced Network Interface (FTE based)	51454196-100
EPDG EPDG-2	Enhanced Peripheral Display Generator	51401286-100 51402089-100
FDC	Floppy Disk Controller	51400669-100
HDDT	Hard Disk Drive Tray	51402176-100
HMPU	High Perf. Module Processor Unit (incl: coprocessor, 2 MB mem)	51400978-100
HPK2-2	High Perf. Module Proc. (incl: 2 MB memory, no coprocessor)	51401288-100
HPK2-3	High Perf. Module Proc. (incl: 3 MB memory, no coprocessor)	51401288-200
K2LCN-x	68020 High Density Kernel (2, 3, 4, 6, 8 MB of memory)	51401551-x00
K2LCN-x	68020 High Density Kernel (2, 4, 8 MB of memory)	51402615-x00
K4LCN-x	68040 High Density Kernel (4, 8, or 16 MB of memory) Separate 4 MB of memory Separate 8 MB of memory Separate 16 MB of memory	51401946-100 51201645-400 51201645-800 51201645-160
K4LCN-x	68040 High Density Kernel (4, 8, or 16 MB of memory) Separate 4 MB of memory Separate 8 MB of memory Separate 16 MB of memory	51402755-100 51201795-400 51201795-800 51201793-160
LCN	Local Control Network Interface	51400667-100
LLCN	Low Power LCN Interface	51401291-100
MMEM	Memory Board, 1 MB	8036211-100
MMEM	Memory Board, 3/4 MB	8036211-200
NGI	Network Gateway Interface	51401583-200
PDG	Peripheral Display Generator Interface	51400926-100
PLCI	Programmable Logic Controller Interface	51400997-100

(Continued)

Table 2-4 — Functional Board Types (Continued)

Board Type	Description	Part Number
PMEM	2 MB Memory	51400903-100
PNI	Process Network Interface	51400955-100
PNM	Process Network Modem	51401163-100
QMEM-x	2 to 4 MB Quad Enhanced Memory (x = 2, 3, or 4 MB)	51401072-x00
SIO	Serial Input Output Interface	51400655-100
SPC	Smart Peripheral Controller	51401052-100
TPDG	Turbo Peripheral Display Generator	51402000-200
TPDGX	Turbo Peripheral Display Generator (high speed)	51402610-200
VDG	Video Display Generator	51400665-100
WSI	Work Station Interface (16 MB memory)	51304791-300
WSI	Work station Interface (32 MB memory)	51304791-400
WSI2	Work Station Interface (64 MHz PA-RISC) memory separate	51402083-100
WSI2	Work Station Interface (100 MHz PA-RISC) memory separate	51402083-200

I/O boards with the letters "CE" in the description are CE compatible designs that provide grounding for shielded cables.

Table 2-5 — I/O Board or Paddle Board Types

Board Type	Description	Part Number
AMR I/O	Redundant AM Interface I/O	51304159-100
AMR I/O	AM Redundancy I/O CE	51304159-200
CLCN-A	LCN I/O Board (Dual Node LCN Coax A I/O Board) CE	51305072-200
CLCN-A	LCN I/O Board (Dual Node LCN Coax A I/O Board) Replaces KLCN-CA	51305072-500
CLCN-A	LCN I/O Board (Dual Node LCN Coax A I/O Board) CE	51305072-800
CLCN-A/B	LCN I/O Board CE	51305072-100
CLCN-A/B	LCN I/O Board	51305072-400
CLCN-A/B	LCN I/O Board CE	51305072-700
CLCN-B	LCN I/O Board (Dual Node LCN Coax B I/O Board) CE	51305072-300
CLCN-B	LCN I/O Board (Dual Node LCN Coax B I/O Board) Replaces KLCN-CB	51305072-600
CLCN-B	LCN I/O Board (Dual Node LCN Coax B I/O Board) CE	51305072-900
CLI I/O	Communications Line Interface I/O (RS-232C)	80360209-001
CLI I/O	Communications Line Interface I/O (RS-449)	80360230-001
CLI/A	Communications Line Interface (RS-449) CE	51196701-100
CNI I/O	Communications Network Interface	51304537-100
CNI I/O	Communications Network Interface I/O Board CE	51304537-200
CS/R	Clock Source/Repeater	51109919-100
CS/R	Clock Source/Repeater (with Precision Clock also called PCS/R)	51304286-100
CS/R	Clock Source/Repeater CE	51304286-200
DHI I/O	Data Hiway Interface I/O	51108088-100
DHI I/O	Data Hiway Interface I/O Board CE	51108088-200
EPNI2 I/O	Enhanced Network Interface I/O	51306816-100
EPNI2 I/O	Enhanced Network Interface I/O CE	51306816-200
EPDG I/O	Enhanced Peripheral Display Generator I/O	51304270-100
EPDGC-1	Enhanced Peripheral Display Generator I/O Board-1 CE	51402447-100
EPDGC-2	Enhanced Peripheral Display Generator I/O Board-2 CE	51402447-200
EPDGP I/O	Enhanced Peripheral Display Generator I/O (replaces EPDG I/O)	51304584-300
EPLCI I/O	Enhanced Programmable Logic Controller Interface I/O	51304812-100
EPLCI I/O	Enhanced Programmable Logic Controller Interface I/O CE	51304812-200

(Continued)

Table 2-5 — I/O Board or Paddle Board Types (Continued)

Board Type	Description	Part Number
FDC I/O	Floppy Disk Controller I/O	51109336-100
HDDT I/O	Hard Disk Drive Tray I/O	51402176-100
HDDT I/O	Hard Disk Drive Tray I/O Board CE	51304913-200
LCN I/O	Local Control Network Interface I/O	51107403-100
XLCNE2	Multi Mode (8Km)	* 51305508-100
XLCNE2	Multi Mode (8Km) CE	* 51305508-200
XLCNE2	Single Mode (20Km)	* 51305508-300 ₁
XLCNE2	Single Mode (20Km) CE	* 51305508-400 ₁
	¹ For systems with XLCNE2s (Single Mode), If the Total Network Length exceeds 10.3 km (for a single remote segment topology) or 9.0 km (for a multiple remote segment topology), modifications to every LCN interface in the LCN system is required. That is, every LLCN board needs to be replaced by an XLLCN board and every K2LCN/K4LCN board needs to be replaced with an XK2LCN/XK4LCN board and for GUS stations the LCNP4 needs to be replaced with the XLCNP4.	
MCPU I/O	Processor “typer” I/O (with clock source and RS-232C interface)	51107754-100
MCPU I/O	Processor “typer” I/O (with RS-232C interface only)	51107954-100
NG FOM	Network Gateway Fiber Optic Modem CE 51306959-200 (redesigned by Honeywell – Active) 2005A CD Networks (Honeywell Special - obsolete)	51306959-200
NGIO	Network Gateway I/O Board	51304472-100
NIM MODEM	NIM Modem I/O board (put in PNI I/O slot)	51304511-100
NIM Modem	NIM Modem I/O CE	51304511-200
PLCG Relay	Programmable Logic Controller Gateway Relay Board	51304421-100
PLCG Relay	Programmable Logic Controller Gateway Relay Board CE	51304421-200
PLCI I/O	Programmable Logic Controller Interface I/O	51195096-100
PLCI I/O	Programmable Logic Controller Interface I/O Board CE	51195096-200
PNI I/O	Process Network Interface I/O	51303944-100
PNM I/O	Process Network Modem I/O	51303947-100

(Continued)

Table 2-5 — I/O Board or Paddle Board Types (Continued)

Board Type	Description	Part Number
SIO I/O	Serial Input Output Interface I/O	51108843-100
SIOI-OE	Serial Input Output I/O Enhanced	51304814-100
SIOI-OE	Serial Input Output I/O Enhanced CE	51304814-200
SPC I/O	Smart Peripheral Controller I/O	51304156-100
SPC-2	Smart Peripheral Controller I/O Board	51304907-100
SPC-2	Smart Peripheral Controller I/O Board CE	51304907-200
SPC-3 I/O	Smart Peripheral Controller I/O Board CE	51305088-100
TP485	TP485 LCN Interface Board	51304776-100
TP485	TP485 LCN Interface Board	51304776-200
TP485-3	TP485 LCN Interface Board CE	51304776-300
TP485-4	TP485 LCN Interface Board CE	51304776-400
TPDG I/O	Turbo Peripheral Display Generator I/O	51304831-200
TPDGC I/O	Turbo Peripheral Display Generator I/O Board CE	51305075-100
WDC I/O ^{W1}	Winchester Disk Controller I/O (for WREN I)	51108678-100
WDC I/O ^{W2}	Winchester Disk Controller I/O (for WREN II)	51108676-100
WDI	Winchester Drive Interface	51304903-100
WDI	Winchester Drive Interface CE	51304903-200
WSI I/O	Work Station Interface I/O	51304791-100
WSI2 I/O	Work Station Interface 2 I/O	51304924-200
WSI2 I/O	Work Station Interface I/O Board CE	51304924-200
WSI2R I/O	Work Station Interface 2 I/O (R100 to R200 retrofit)	51304947-100

Older board types (especially memory boards) may be retained while upgrading to the R500 software release, provided they meet the constraints listed in Table 2-6 and/or your particular configuration listed in the tables found in subsections 2.2.2 through 2.2.26.

NOTE

K2LCN and K4LCN-4 boards will not function with external memory. External memory is never addressed.

Table 2-6 — Processor/Memory Board Compatibility Matrix

Memory Board Type	Processor Type		
	EMPU (68000 CPU)	HMPU (68020 CPU, 2 MB, with Coprocessor)	HPK2-2 <i>Note 4</i> or HPK2-3 (68020 CPU with 2 or 3 MB Memory)
MEM—1/4 MEG 51400659-100	YES <i>Note 1</i>	NO	NO
MMEM—3/4 MEG 80360211-200	YES <i>Note 1</i>	NO	NO
MMEM—1 MEG 580360211-100	YES <i>Note 1</i>	NO	NO
EMEM—1 MEG 51400910-100	YES	YES (EMEM must be at Revision E)	YES (EMEM must be at Revision E)
PMEM—2 MEG 51400903-100	YES <i>Note 2</i>	YES	NO
QMEM-1—1 MEG (WITHDRAWN)	NO	NO	NO
QMEM-2, -3, -4 2 through 4 MEG 51401072-x00 (x=2,3,4 for 2,3,4 MB)	YES <i>Note 3</i>	YES <i>Note 3</i>	YES <i>Note 3</i>

Note 1 YES, but must have one or more EMEMs (or QMEM-1) in lowest-numbered memory slots.

Note 2 Use PMEM only in Upper Tier US or Universal Work Station—don't use for control.

Note 3 Place QMEM in lowest-numbered memory slot—don't use with MEM or MMEM boards.

Note 4 All HPK2-2 and HPK2-3 boards must be at Revision G or greater.

* CLCN A/B is the CE replacement for the LCN I/O board.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.2 Application Module Configurations

Table 2-7 — Application Module (AM) R500 Factory Configurations

Type	P-MAD15-100		P-MAD16-200		P-MAD17-200	
Slot	Front	Rear	Front	Rear	Front	Rear
5						
4						
3			EMEM		QMEM-2	
2 <i>Note 7</i>	LLCN	LCN I/O	LLCN	LCN I/O	AMR	AMR I/O
1 <i>Note 5</i>	HPK2-3	<i>Note 6</i>	HPK2-3	<i>Note 6</i>	LLCN	LCN I/O
					HMPU	<i>Note 6</i>

Type	P-MAD19-100 (Release 321 or later)		AM Redundancy	
Slot	Front	Rear	Front	Rear
5	QMEM-3			
4	QMEM-2		QMEM-1	
3	AMR	AMR I/O	AMR	AMR I/O
2 <i>Note 7</i>	LLCN	LCN I/O	LLCN	LCN I/O
1 <i>Note 5</i>	HMPU	<i>Note 6</i>	HMPU	<i>Note 6</i>

Note 5 HMPU and HPK2-2 boards are interchangeable except in a redundant AM.

Note 6 Optional clock source/typer boards CS/R or MCPU I/O (see Table 2-5) may be included with any module type. Clock sources are generally placed in either the HG or NIM module.

Note 7 LCN and LLCN (Low Power LCN) boards are interchangeable.

Application Modules that are upgraded from previous releases with upgrade kit MP-PROCx (K2LCN), MP-ZPRCxx (K4LCN), or MP-ZPSCxx (K4LCN) will have one of the following board configurations.

Table 2-8 — Redundant Application Modules R500 Upgraded Configurations

Type	3 Megaword		4 Megaword		5 Megaword	
Slot	Front	Rear	Front	Rear	Front	Rear
5						
4						
3						
2	EAMR	AMR I/O	EAMR	AMR I/O	EAMR	AMR I/O
1	K4LCN-4	LCN I/O	K4LCN-4	LCN I/O	K4LCN-8	LCN I/O

Type	6 Megaword		7 Megaword		8 Megaword	
Slot	Front	Rear	Front	Rear	Front	Rear
5						
4						
3						
2	EAMR	AMR I/O	EAMR	AMR I/O	EAMR	AMR I/O
1	K4LCN-8	LCN I/O	K4LCN-8	LCN I/O	K4LCN-8	LCN I/O

Table 2-9 — Non-Redundant Application Modules R500 Upgraded Configurations

Type	3 Megaword		4 Megaword		5 Megaword	
Slot	Front	Rear	Front	Rear	Front	Rear
5						
4						
3						
2						
1	K2LCN-3 or K4LCN-4	LCN I/O	K2LCN-3 or K4LCN-4	LCN I/O	K2LCN-6 or K4LCN-8	LCN I/O

Type	6 Megaword		7 Megaword		8 Megaword	
Slot	Front	Rear	Front	Rear	Front	Rear
5						
4						
3						
2						
1	K2LCN-6 or K4LCN-8	LCN I/O	K2LCN-8 or K4LCN-8	LCN I/O	K2LCN-8 or K4LCN-8	LCN I/O

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.3 Application Module Configurations (CE Compliant)

Release 500 Application Modules are built in Dual Nodule Module versions only.

Table 2-10 — Redundant Application Module (AM) R500 Factory Configurations—Five-Slot

Type	MP-AMDC34-100 MP-AMDC34-200 MP-AMDC34-300		MP-AMDC44-100 MP-AMDC44-200 MP-AMDC44-300		MP-AMDH46-100 MP-AMDH46-200 MP-AMDH46-110	
Slot	Front	Rear	Front	Rear	Front	Rear
5						
4						
3	QMEM-2		QMEM-2		QMEM-4	
2 <i>Note 7</i>	LLCN	CLCN	LLCN	CLCN	LLCN	CLCN
1 <i>Note 5</i>	HMPU	<i>Note 6</i>	HMPU	<i>Note 6</i>	HMPU	<i>Note 6</i>

Type	MP-AMDR13-100 MP-AMDR13-200		MP-AMDR14-100 MP-AMDR14-200		MP-AMDR16-100 MP-AMDR16-200	
Slot	Front	Rear	Front	Rear	Front	Rear
5			QMEM-2			
4	QMEM-4		QMEM-3		QMEM-4	
3	AMR	AMR	AMR	AMR	AMR	AMR
2 <i>Note 7</i>	LLCN	CLCN	LLCN	CLCN	LLCN	CLCN
1 <i>Note 5</i>	HMPU	<i>Note 6</i>	HMPU	<i>Note 6</i>	HMPU	<i>Note 6</i>

Type	MP-AMDR17-100 MP-AMDR17-200		Reserved		Reserved	
Slot	Front	Rear				
5	QMEM-2					
4	QMEM-3					
3	AMR	AMR				
2 <i>Note 7</i>	LLCN	CLCN				
1 <i>Note 5</i>	HMPU	<i>Note 6</i>				

Note 5 HMPU and HPK2-2 boards are interchangeable except in a redundant AM.

Note 6 Optional clock source/typer boards CS/R (see Table 2-5) may be included with any module type. Clock sources are generally placed in either the HG or NIM module.

Note 7 LCN and LLCN (Low Power LCN) boards are interchangeable.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

Table 2-11 — Redundant Application Module (AM) R500 Factory Configurations—Five-Slot

Type	MP-AMDR13-100 MP-AMDR13-200 2 Mw		MP-AMDR14-100 MP-AMDR14-200		MP-AMDR16-100 MP-AMDR16-200 4 Mw	
Slot	Front	Rear	Front	Rear	Front	Rear
5			EMEM			
4	QMEM-2		EMEM		QMEM-4	
3	AMR	AMR (I/O)	AMR	AMR (I/O)	AMR	AMR (I/O)
2 <i>Note 7</i>	LLCN	CLCN	LLCN	CLCN	LLCN	CLCN A/B
1 <i>Note 5</i>	HMPU	<i>Note 6</i>	HMPU	<i>Note 6</i>	HMPU	<i>Note 6</i>

Type	MP-AMDR17-100 MP-AMDR17-200 7 Mw		Reserved		Reserved	
Slot	Front	Rear				
5						
4	QMEM-4					
3	AMR	AMR (I/O)				
2 <i>Note 7</i>	LLCN	CLCN A/B				
1 <i>Note 5</i>	HMPU	<i>Note 6</i>				

Note 5 HMPU and HPK2-2 boards are interchangeable except in the Redundant AM.

Note 6 Optional clock source/typer boards CS/R (see Table 2-5) may be included with any module type. Clock sources are generally placed in either the HG or NIM module.

Note 7 LCN and LLCN (Low Power LCN) boards are interchangeable.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.4 Application Module^X Configurations

2.2.4

Application Module^X software releases are controlled by a separate release system and are "release independent" of LCN software releases. Refer to *Application Module^X Service* manual.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.5 Computer Gateway Configurations

A 2-megaword Computer Gateway that is upgraded from previous releases with upgrade kit MP-PROCx (K2LCN), MP-ZPRCxx (K4LCN), or MP-ZPSCxx (K4LCN) will have the following board configuration.

Table 2-12 — Computer Gateway (CG) R500 Configurations (Non-CE Compliant)

Type	2 Megaword		P-MCE12-200	
Slot	Front	Rear	Front	Rear
5	CLI	CLI/A I/O**	CLI	CLI/A I/O**
4				
3				
2				
Note 7				
1	K2LCN-2 or K4LCN	LCN I/O	LLCN	LCN I/O
Note 5,				
Note 8				
			HPK2-2	Note 6

Note 5 HMPU and HPK2-2 boards are interchangeable except in the Redundant AM.

Note 6 Optional clock source/typer boards CS/R (see Table 2-5) may be included with any module type. Clock sources are generally placed in either the HG or NIM module.

Note 7 LCN and LLCN (Low Power LCN) boards are interchangeable.

Note 8 Any 5/10 card chassis MUST have the K2/K4 processor board installed in slot 1 unless the node contains a CSR I/O paddle card in slot 1 (which is the LCN coax shield ground for THAT cable). Once the LCN coax shield ground(s) have been installed correctly (refer to the Dual Node Module Service manual (LC13-610), section 2.8 LCN Network Grounding), relocate the K2/K4 and its associated LCN I/O paddle card into slot 1. If the K2/K4 is NOT in slot 1, the system will NOT perform "Backplane Integrity" checks and you could have a hardware problem with the node and not know it.

** CLI/A I/O is the CE replacement for the CLI I/O board.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.6 Computer Gateway Configurations (CE Compliant)

This module is only available in a Dual Node Module for R500 factory built configurations. Refer to the *Dual Node Module Service* manual.

2.2.7 Hiway Gateway Configurations

A 2-megaword Hiway Gateway that is upgraded from previous releases with upgrade kit MP-PROCx (K2LCN), MP-ZPRCxx (K4LCN), or MP-ZPSCxx (K4LCN) will have the following board configuration.

Table 2-13 — Hiway Gateway (HG) R500 Configurations (Non-CE Compliant)

Type	2 Megaword		P-MHG13-100	
	Front	Rear	Front	Rear
5				
4	DHI	DHI I/O	DHI	DHI I/O
3				
2 <i>Note 7</i>	K2LCN-2 or K4LCN	LCN I/O	LLCN	LCN I/O
1 <i>Note 5, Note 8</i>		<i>Note 6</i> CSR I/O	HPK2-2	

Note 5 HMPU and HPK2-2 boards are interchangeable except in the Redundant AM.

Note 6 Optional clock source/typer boards CS/R (see Table 2-5) may be included with any module type. Clock sources are generally placed in either the HG or NIM module.

Note 7 LCN and LLCN (Low Power LCN) boards are interchangeable.

Note 8 Any 5/10 card chassis MUST have the K2/K4 processor board installed in slot 1 unless the node contains a CSR I/O paddle card in slot 1 (which is the LCN coax shield ground for THAT cable). Once the LCN coax shield ground(s) have been installed correctly (refer to the Dual Node Module Service manual (LC13-610), section 2.8 LCN Network Grounding), relocate the K2/K4 and its associated LCN I/O paddle card into slot 1. If the K2/K4 is NOT in slot 1, the system will NOT perform "Backplane Integrity" checks and you could have a hardware problem with the node and not know it.

2.2.8 Hiway Gateway Configurations (CE Compliant)

This module is only available in a Dual Node Module for R500 factory built configurations. Refer to the *Dual Node Module Service* manual.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.9 History Module Configurations

History Modules that are upgraded from previous releases with upgrade kit MP-PROCx (K2LCN), MP-ZPRCxx (K4LCN), or MP-ZPSCxx (K4LCN) will have one of the following board configurations.

Table 2-14 — History Module (HM) R500 Upgraded Configurations Non-CE Compliant (WREN III)

Type	3 megaword		3 megaword	
Slot	Front	Rear	Front	Rear
5	SPC	SPC I/O	SPC	SPC I/O
4				
3			EMEM	
2 <i>Note 7</i>	K2LCN-3 or K4LCN	LCN I/O	LLCN	LCN I/O
1 <i>Note 5, Note 8</i>		<i>Note 6</i> CSR I/O	HPK2-2	

Table 2-15 — WDA History Module (HM) R500 Factory Configurations (CE Compliant)

Type	3 megaword		4 megaword	
Slot	Front	Rear	Front	Rear
5				
4	WDA	WDI	WDA	WDI
3				
2	SPC	SPC-2	SPC	SPC-2
1	K2LCN-3	LCN I/O	K4LCN-4	LCN I/O

Note 5 HMPU and HPK2-2 boards are interchangeable except in the Redundant AM.

Note 6 Optional clock source/typer boards CS/R or MCPU I/O (see Table 2-5) may be included with any module type. Clock sources are generally placed in either the HG or NIM module.

Note 7 LCN and LLCN (Low Power LCN) boards are interchangeable.

Note xx Winchester Drive Assembly (WDA) require two slots.

Note 8 Any 5/10 card chassis MUST have the K2/K4 processor board installed in slot 1 unless the node contains a CSR I/O paddle card in slot 1 (which is the LCN coax shield ground for THAT cable). Once the LCN coax shield ground(s) have been installed correctly (refer to the Dual Node Module Service manual (LC13-610), section 2.8 LCN Network Grounding), relocate the K2/K4 and its associated LCN I/O paddle card into slot 1. If the K2/K4 is NOT in slot 1, the system will NOT perform "Backplane Integrity" checks and you could have a hardware problem with the node and not know it.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.10 History Module Configurations (CE Compliant)

Table 2-16 — History Module (HM) R500 Factory Configurations (Winchester Disk Assembly)

Type	MP-HMSD84-100 MP-HMSD84-200 MP-HMSD84-300 MP-HMSR84-100 MP-HMSR84-200 MP-HMSR84-300 MP-HMDD84-100 MP-HMDD84-200 MP-HMDD84-300 MP-HMDR84-100 MP-HMDR84-200 MP-HMDR84-300		MP-HMSD92-100 MP-HMSD92-200 MP-HMSD92-300 MP-HMDD92-100 MP-HMDD92-200 MP-HMDD92-300 MP-HMSR92-100 MP-HMSR92-200 MP-HMSR92-300 MP-HMDR92-100 MP-HMDR92-200 MP-HMDR92-300	
Slot	Front	Rear	Front	Rear
5				
4	WDA	WDI ***	WDA	WDI ***
3				
2	SPC	SPC2 I/O**	SPC	SPC2 I/O**
1	K2LCN-3	CLCN A/B*	K4LCN-4	CLCN A/B*

* CLCN A/B is the CE replacement for the LCN I/O board.

** SPC-2 I/O is the CE replacement for the SPC I/O board.

*** WDI is the CE replacement for the WDI I/O board.

Note xx Winchester Drive Assemblies (WDA) require two slots.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.11 Network Gateway Configurations

A Network Gateway that is upgraded from previous releases with upgrade kit MP-PROCx (K2LCN), MP-ZPRCxx (K4LCN), or MP-ZPSCxx (K4LCN) will have the following board configuration.

Table 2-17 — Network Gateway (NG) R500 Upgraded Configuration (Non CE Compliant)

Type	High Performance Only	
	Front	Rear
5		
4		
3 <i>Note 14</i>	NGI	NGI ** or NG FOM***
2	NGI	NGI ** or NG FOM***
1*	K2LCN-2 or K4LCN	LCN I/O

Note 14 The NGI and NGIO in Slot 3 are optional. They are present for a redundant cable A & B NG communications.

* Any 5/10 card chassis **MUST** have the K2/K4 processor board installed in slot 1 unless the node contains a CSR I/O paddle card in slot 1 (which is the LCN coax shield ground for THAT cable). Once the LCN coax shield ground(s) have been installed correctly (refer to the Dual Node Module Service manual (LC13-610), section 2.8 LCN Network Grounding), relocate the K2/K4 and its associated LCN I/O paddle card into slot 1. If the K2/K4 is NOT in slot 1, the system will NOT perform "Backplane Integrity" checks and you could have a hardware problem with the node and not know it.

** NGI is the CE replacement for the NGI I/O board.

*** NG FOM is the CE replacement for the NG fiber optic I/O board.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, except for the CLCNA I/O, CLCNCB I/O, KLCNA I/O, and or KLCNCB I/O paddle cards. These boards are the ONLY boards that can be inserted with power on. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.12 Network Interface Module Configurations

2.2.12

The PNM, PNM I/O, and PNI I/O boards and the mini-coax can be replaced by the NIM MODEM board. This is done by installing the NIM MODEM Field Replacement Kit, 51195760. The new NIM MODEM board is shown as an alternate configuration in Tables 2-18 and 2-19 below.

Table 2-18 — Network Interface Module (NIM) R500 Factory Configurations

Type	P-MNIM1-200		Alternate Configuration		Alternate Configuration	
Slot	Front	Rear	Front	Rear	Front	Rear
5	PNM	PNM I/O	PNM	PNM I/O	EPNI	NIM Modem
4	PNI	PNI I/O	PNI	PNI I/O		
3	EMEM	LCN I/O	LLCN	LCN I/O	LLCN	LCN I/O
2 <i>Note 7</i>	LLCN					
1	HPK2-2					

Table 2-19 — Network Interface Module (NIM) R500 Upgraded Configurations

Type	High Performance Only		Alternate Configuration	
Slot	Front	Rear	Front	Rear
5	PNM	PNM I/O	EPNI	NIM Modem
4	PNI	PNI I/O		
3	EMEM	LCN I/O	EMEM	LCN I/O
2 <i>Note 7</i>	LCN		LCN	
1 <i>Note 5</i>	HMPU		HMPU	

Table 2-20— Network Interface Module (NIM) R500 Configurations (Non-CE Compliant)

Type	2 Megaword		P-MNIM1-200	
Slot	Front	Rear	Front	Rear
5	EPNI	NIM Modem	EPNI	NIM Modem
4 <i>Note 9</i>				
3	K2LCN-2 or K4LCN	LCN I/O	K2LCN-2 or K4LCN	LCN I/O
2				
1 <i>Note 8</i>		CSR I/O <i>Note 6</i>	HPK2-2 <i>Note 5</i>	

Note 5 HMPU and HPK2-2 boards are interchangeable except in the Redundant AM.

Note 6 Optional clock source/typer boards CS/R (see Table 2-5) may be included with any module type. Clock sources are generally placed in either the HG or NIM module.

Note 7 LCN and LLCN (Low Power LCN) boards are interchangeable.

Note 8 Any 5/10 card chassis MUST have the K2/K4 processor board installed in slot 1 unless the node contains a CSR I/O paddle card in slot 1 (which is the LCN coax shield ground for THAT cable). Once the LCN coax shield ground(s) have been installed correctly (refer to the Dual Node Module Service manual (LC13-610), section 2.8 LCN Network Grounding), relocate the K2/K4 and its associated LCN I/O paddle card into slot 1. If the K2/K4 is NOT in slot 1, the system will NOT perform "Backplane Integrity" checks and you could have a hardware problem with the node and not know it.

Note 9 EPNI is a newer version of PNI.

2.2.13 Network Interface Module Configurations (CE Compliant) 2.2.13

This module is only available in a Dual Node Module. Refer to the *Dual Node Module Service* manual.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.14 Enhanced Network Interface Module Configurations

You can replace the NIM MODEM with the EPNI2 I/O board. This is performed by installing the ENIM Upgrade Kit, MP-ZENIMR.

The new EPNI2 I/O board configuration is shown in the following Table 2-21:

Table 2-21 — Enhanced Network Interface Module (ENIM) Configurations

Type	ENIM		Alternate Configuration	
Slot	Front	Rear	Front	Rear
5	EPNI2 LCN <i>Note 18a and Note 18b</i>	EPNI2 I/O LCN I/O	EPNI2 K2LCN-3 or K4LCN <i>Note 18a and Note 18b</i>	EPNI2 I/O
4				
3				
2				
1				

Note 18a ENIM does not support LCN boards of HMPU and HPK2 types.

Note 18b The minimum supported LCN board for ENIM is K2LCN-3.

A 2-megaword PLC Gateway that is upgraded from previous releases with upgrade kit MP-PROCx (K2LCN), MP-ZPRCxx (K4LCN), or MP-ZPSCxx (K4LCN) will have the following board configuration.

Table 2-22 — PLC Gateway (PLCG) R500 Upgraded Configurations (Non-CE Compliant)

Type	2 Megaword		P-MLCG3-100	
Slot	Front	Rear	Front	Rear
5				
4	PLCI/EPLCI	PLCI I/O**	PLCI/EPLCI	PLCI I/O**
3				
2 <i>Note 7</i>	K2LCN-2 or K4LCN	LCN I/O	LLCN	LCN I/O
1 <i>Note 5, Note 8</i>		<i>Note 6</i> CSR I/O	HPK2-2	<i>Note 6</i> CSR I/O

Note 5 HMPU and HPK2-2 boards are interchangeable except in the Redundant AM.

Note 6 Optional clock source/typer boards CS/R or MCPU I/O (see Table 2-5) may be included with any module type. Clock sources are generally placed in either the HG or NIM module.

Note 7 LCN and LLCN (Low Power LCN) boards are interchangeable.

Note 8 Any 5/10 card chassis MUST have the K2/K4 processor board installed in slot 1 unless the node contains a CSR I/O paddle card in slot 1 (which is the LCN coax shield ground for THAT cable). Once the LCN coax shield ground(s) have been installed correctly (refer to the Dual Node Module Service manual (LC13-610), section 2.8 LCN Network Grounding), relocate the K2/K4 and its associated LCN I/O paddle card into slot 1. If the K2/K4 is NOT in slot 1, the system will NOT perform "Backplane Integrity" checks and you could have a hardware problem with the node and not know it.

** PLCI is the CE replacement for the PLCI I/O.

This module is only available in a Dual Node Module. Refer to the *Dual Node Module Service* manual.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.17 Plant Network Module Configurations

Table 2-23 — Plant Network Module (PLNM) R500 Factory Configurations

Type	High Performance	
Slot	Front	Rear
5	CNI	CNI I/O
4		
3		
2 <i>Note 7</i>	LLCN	LCN I/O
1	HPK2-2	

Note 7 LCN and LLCN (Low Power LCN) boards are interchangeable.

2.2.18 Plant Network Module Configurations (CE Compliant)

This module is only available only in a Dual Node Module. Refer to the *Dual Node Module Service* manual.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.19 Universal Station Configurations

Universal Stations that are upgraded from previous releases with upgrade kit MP-PROCx (K2LCN), MP-ZPRCxx (K4LCN), or MP-ZPSCxx (K4LCN) will have one of the following board configurations.

A US with an Operator personality requires 4 Mw of memory. A US with a Universal personality requires 6 Mw of memory.

Table 2-24 — Universal Station (US) R500 Upgraded Configurations (Non CE Compliant)

Type	4 Mw (Operator)		6 Mw (Universal)	
Slot	Front	Rear	Front	Rear
5	EPDG	EPDG I/O	EPDG	EPDG I/O
4				
3				
2	K2LCN-4 or K4LCN-4	LCN I/O	K2LCN-6 or K4LCN-8	LCN I/O
1		Note 6 CSR I/O		Note 6 CSR I/O

Table 2-25 — Universal Station (US) R500 Upgraded Configurations (Non CE Compliant)

Type	4 Mw (Operator)		6 Mw (Universal)	
Slot	Front	Rear		
10				
9				
8				
7				
6 Note 11	SIO	SIO I/O	SIO	SIO I/O
5 Note 10	(Do Not Use)		(Do Not Use)	
4	EPDG	EPDG I/O	EPDG	EPDGP I/O
3 Note 9	FDC	FDC I/O	FDC	FDC I/O
2	K2LCN-4 or K4LCN-4	LCN I/O	K2LCN-6 or K4LCN-8	LCN I/O
1 Note 12		Note 6 CSR I/O		Note 6 CSR I/O

Note 6 Optional clock source/typer boards CS/R or MCPU I/O (see Table 2-5) may be included with any module type. Clock sources are generally placed in either the HG or NIM module.

Note 9 These slots may contain optional FDC & FDC I/O or SPC & SPC I/O boards.

Note 10 Do not use the slot above EPDG—use only certain boards in the 2nd slot above EPDG.

Note 11 SIO & SIO I/O boards are optional.

Note 12 Any 5/10 card chassis MUST have the K2/K4 processor board installed in slot 1 unless the node contains a CSR I/O paddle card in slot 1 (which is the LCN coax shield ground for THAT cable). Once the LCN coax shield ground(s) have been installed correctly (refer to the Dual Node Module Service manual (LC13-610), section 2.8 LCN Network Grounding), relocate the K2/K4 and its associated LCN I/O paddle card into slot 1. If the K2/K4 is NOT in slot 1, the system will NOT perform “Backplane Integrity” checks and you could have a hardware problem with the node and not know it.

2.2.20 Universal Station Configurations (CE Compliant)

This module is available only in a Dual Node Module. Refer to the *Dual Node Module Service* manual.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.21 Universal Station^X Configurations

Table 2-26 — Universal Station^X (U^XS) R500 Factory R100 Configurations

Type	MP-MMUSX1, 2, 3, 4	
Slot	Front	Rear
5	TPDGX	TPDG I/O
4		
3		
2 <i>Note 16</i>	WSI	WSI I/O
1	K2LCN-8	LCN I/O

Table 2-27 — Universal Station^X (U^XS) R500 Upgraded R200 Configurations

Type	Upgrade Kit MP-ZRSKUG Installed		Upgrade Kit MP-ZUXCC2 or MP-ZUXNC2 Installed		Upgrade Kit MP-MUSX13, 14, 15, 16 Installed	
Slot	Front	Rear	Front	Rear	Front	Rear
5	TPDGX	TPDG I/O	TPDGX	TPDG I/O	TPDGX	TPDG I/O
4						
3			HDDT	HDDT I/O	HDDT	HDDT I/O
2	WSI2	WSI2R I/O	WSI2	WSI2 I/O	WSI2	WSI2 I/O
1	K2LCN-8	LCN I/O	K2LCN-8	LCN I/O	K2LCN-8	LCN I/O

Type	MP-MMUSX1, 2, 3, 4		Upgrade Kit MP-ZUXCC2 or MP-ZUXNC2 Installed		Upgrade Kit MP-MUSX13, 14, 15, 16 Installed	
Slot	Front	Rear	Front	Rear	Front	Rear
5	TPDGX	TPDG I/O	TPDGX	TPDG I/O	TPDGX	TPDG I/O
4						
3			HDDT	HDDT I/O	HDDT	HDDT I/O
2	WSI2	WSI2R I/O	WSI2	WSI2 I/O	WSI2	WSI2 I/O
1	K2LCN-8	LCN I/O	K2LCN-8	LCN I/O	K2LCN-8	LCN I/O

Note 16 Use 51402030-100 for 16 MB Work Station Interface (WSI) which is the only setting Honeywell supports.

Note 17 Any 5/10 card chassis MUST have the K2/K4 processor board installed in slot 1 unless the node contains a CSR I/O paddle card in slot 1 (which is the LCN coax shield ground for THAT cable). Once the LCN coax shield ground(s) have been installed correctly (refer to the Dual Node Module Service manual (LC13-610), section 2.8 LCN Network Grounding), relocate the K2/K4 and its associated LCN I/O paddle card into slot 1. If the K2/K4 is NOT in slot 1, the system will NOT perform "Backplane Integrity" checks and you could have a hardware problem with the node and not know it.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.22 Universal Station^X Configurations (CE Compliant)

Table 2-28 —WSI2 Board Assemblies Applications

Model Number	Operation	Graphics RAM for 2nd Monitor	Part Number
MP-MUSX33 (Classic) MP-MUSX35 (Ergonomic)	64 MHz	No	51402083-100
MP-MUSX41 (Classic) MP-MUSX43 (Ergonomic)	100 MHz	No	51402083-200
MP-MUSX34 (Classic) MP-MUSX36 (Ergonomic)	64 MHz	YES	51402083-300
MP-MUSX42 (Classic) MP-MUSX44 (Ergonomic)	100 MHz	YES	51402083-400

Table 2-29— Universal Station^X (U^XS) R500 Factory Configurations (CE Compliant)

Type	MP-MUSX33 ,34,35,36,41,42	
Slot	Front	Rear
5	TPDGX	TPDGC
4		
3	HDDT	HDDT
2 <i>Note 15</i>	WSI2	WSI2
1	K2LCN-8	CLCN A/B*

Note 15 Two versions of the WSI2 board are available to provide either the 64 MHz (minimum configuration) or 100 MHz (high-level configuration) coprocessor. Honeywell only supports the 32 bit.

* CLCN A/B is the CE replacement for the LCN I/O board.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.23 Universal Work Station Configurations

Universal Work Stations that are upgraded from previous releases with upgrade kit MP-PROCx (K2LCN), MP-ZPRCxx (K4LCN), or MP-ZPSCxx (K4LCN) will have one of the following board configurations.

Table 2-30 — Universal Work Station R500 Upgraded Configurations (Non-CE Compliant)

Type	4 Mw (Operator)		6 Mw (Universal)	
Slot	Front	Rear	Front	Rear
5	EPDG	EPDG I/O	EPDG	EPDG I/O
4				
3				
2				
1	K2LCN-4 or K4LCN-4	LCN I/O	K2LCN-6 or K4LCN-8	LCN I/O

2.2.24 Universal Work Station Configurations (CE Compliant)

Table 2-31 — Universal Work Station R500 Factory Configurations (CE Compliant)

Type	MP-UWSM07 4 Mw (Operator) (68040)		MP-UWSM07 8 Mw (Universal) (68040)	
Slot	Front	Rear	Front	Rear
5	EPDG	EPDGC I/O	EPDG	EPDGC I/O
4				
3				
2				
1	K4LCN-4	CLCNC A/B*	K4LCN-8	CLCNC A/B*

* CLCN A/B is the CE replacement for the LCN I/O board.

CAUTION

Power must be removed from the module whenever you are removing or installing any board, including an I/O paddle board. Be sure that an I/O paddle board is installed in the correct slot; some boards have only one slot that they can be installed in without causing damage. I/O paddle boards plugged into the wrong slot can cause traces on the backplane to burn open.

2.2.25 Scanner Application Module Configurations

Table 2-32 — Scanner Application Module (SAM) R500 Upgraded Configurations (Non-CE)

Type	QC-SAM101	
	Front	Rear
5	CCP-4	CLM I/O
4		
3	QMEM-4	LCN I/O
2	LLCN	
1	HMPU	

2.2.26 Scanner Application Module Configurations (CE Compliant)

Information is not available at this time.

2.3 FRONT PANEL

Controls on the front panel consist of a POWER switch, a RESET button, and a MARGIN switch or jumper. The function and operation of the POWER and RESET controls are discussed elsewhere in this manual. The MARGIN switch or pin jumper is a power supply test/maintenance diagnostic aid and should be left in the NOM position at all times.

The front panel contains indicators that monitor module performance and serve as an aid in fault isolation. The indicators consist of light emitting diodes (LED) and a 3-digit alphanumeric display. The LED indicators on the bottom left of the front panel give an indication of the power supply status and an indicator on the right center of the panel lights if a fan module fails. LEDs on each of the boards are used in conjunction with the alphanumeric display to isolate malfunctions on the boards. Further information on the use of the module indicators is located in Section 3 of this manual.

2.4 REAR PANEL

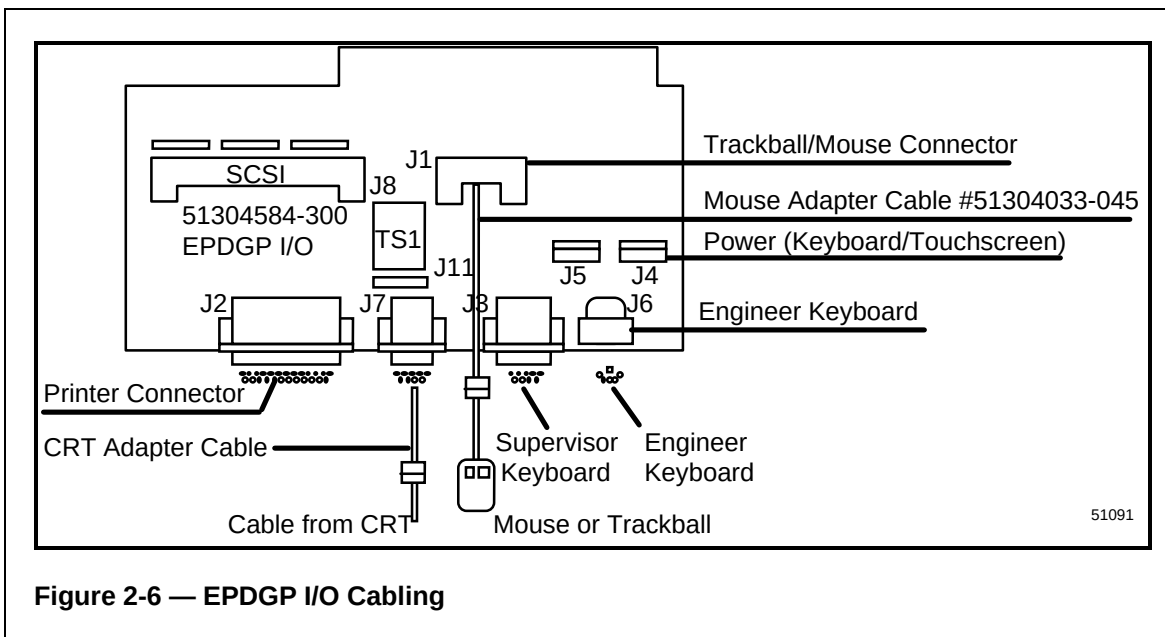
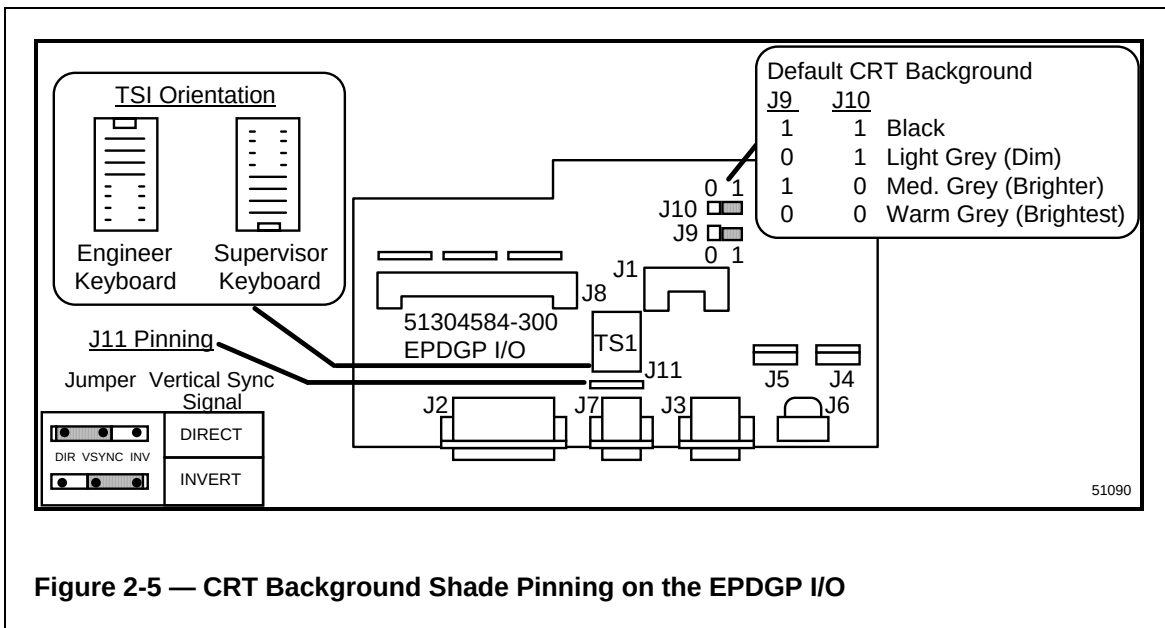
The rear panel contains the I/O board chassis power cable, a 100-pin backplane breakout board, and a grounding lug. As shown in Tables 2-2 and 2-3, the I/O boards are installed in the chassis in the slot corresponding in number to the applicable board installed in the front of the module. All communication with the LCN or the Data Hiway is through the I/O boards. The coaxial cables that run to the boards are connected by a tee connector with the output side of the tee going to the next board (or to a terminating load on the last tee in a series). The I/O board coax connectors are marked A and B; make sure that the A cable connects to the A connector and that the B cable is connected to the B connector. Ribbon cables are used to connect to such items as the Winchester drive module. Other connectors, for example RS-232C or RS-449 on the Computer Gateway, are also used.

2.5 FIELD ADJUSTMENTS

There are no field adjustments for the module. The LCN I/O (CLCN A/B for CE Compliant) board, however, has a module address jumper block that must be characterized for the particular node address it occupies on the LCN. Refer to subsection 8.1 of the *LCN System Installation Manual* for system pinning.

2.6 EPDGP I/O BOARD PINNING

The EPDGP I/O board, if present, has pinning options to set the default background shade for the CRT if a palette has not been set in a schematic (Set Palette is a new command in Release 320). You may find additional information on the Set Palette command in the *Picture Editor Reference Manual*. The EPDGP also has a configuration option that is set for either the Engineer's Keyboard or the Supervisor's Keyboard. (If both keyboards are installed, the EPDGP is set up for the Supervisor's Keyboard, and the Engineer's Keyboard is connected into the Supervisor's Keyboard.) Figure 2-6 shows the keyboard and CRT background options for the EPDGP I/O.



EPDGP I/O boards can be pinned for default background shading as shown in Figure 2-5.

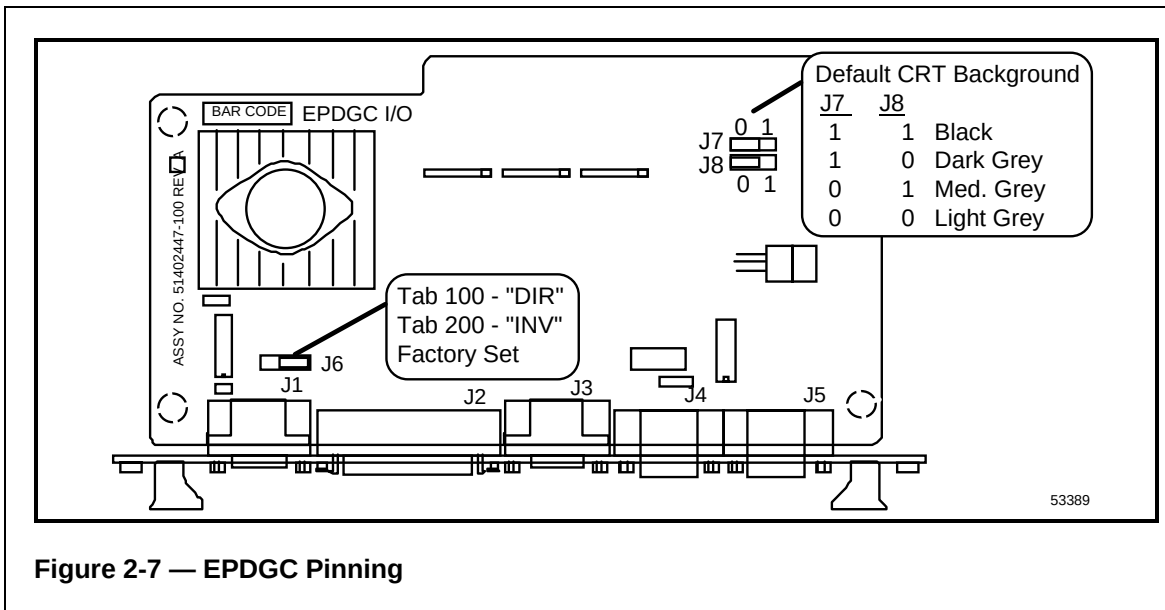
Vertical sync for the monitor is pinned as shown below.

Universal Station Type	EPDGP I/O Assembly
Classic with keyboard power supply	51304584-100
Classic without keyboard power supply	51304584-200
Ergonomic furniture with 21" FST monitor	51304584-300

2.7 EPDGC I/O BOARD PINNING (CE Compliant)

The EPDGC board is the CE Compliant version of the EPDGP I/O board.

The EPDGC board, if present, has pinning options to set the default background shade for the CRT if a palette has not been set in a schematic. You can find additional information on the Set Palette command in the *Picture Editor Reference Manual*. Figure 2-7 shows the CRT background options for the EPDGC I/O board.



2.8 LCN I/O BOARD NODE ADDRESS PINNING

There are no LCN module adjustments; however, the LCN I/O boards have an LCN node address pinning block that must be configured for the particular node it occupies on the LCN. This normally is done for each system during factory test.

The address of the nodes of an LCN must start with "1" and be assigned in increasing numerical order. In planning for future additions to the system, reserve a number, or set of numbers, for the expansion. All other addresses are assigned in the order of module placement. Nodes can be added during system expansion, and can be done without changing the original node assignments if gaps in the node numbering are reserved for nodes that will be added later. A topology map should be initiated and maintained.

If the LCN I/O, CLCNA/B, K2LCN, or K4LCN board is replaced on site, or if an additional module is added, the new board or module (including the Universal Work Station) must be configured for the LCN node that it is assigned. Figures 2-8 through 2-13 illustrate the location of the LCN node address pinning block on the LCN I/O, CLCNA/B, K2LCN, and K4LCN boards.

CAUTION

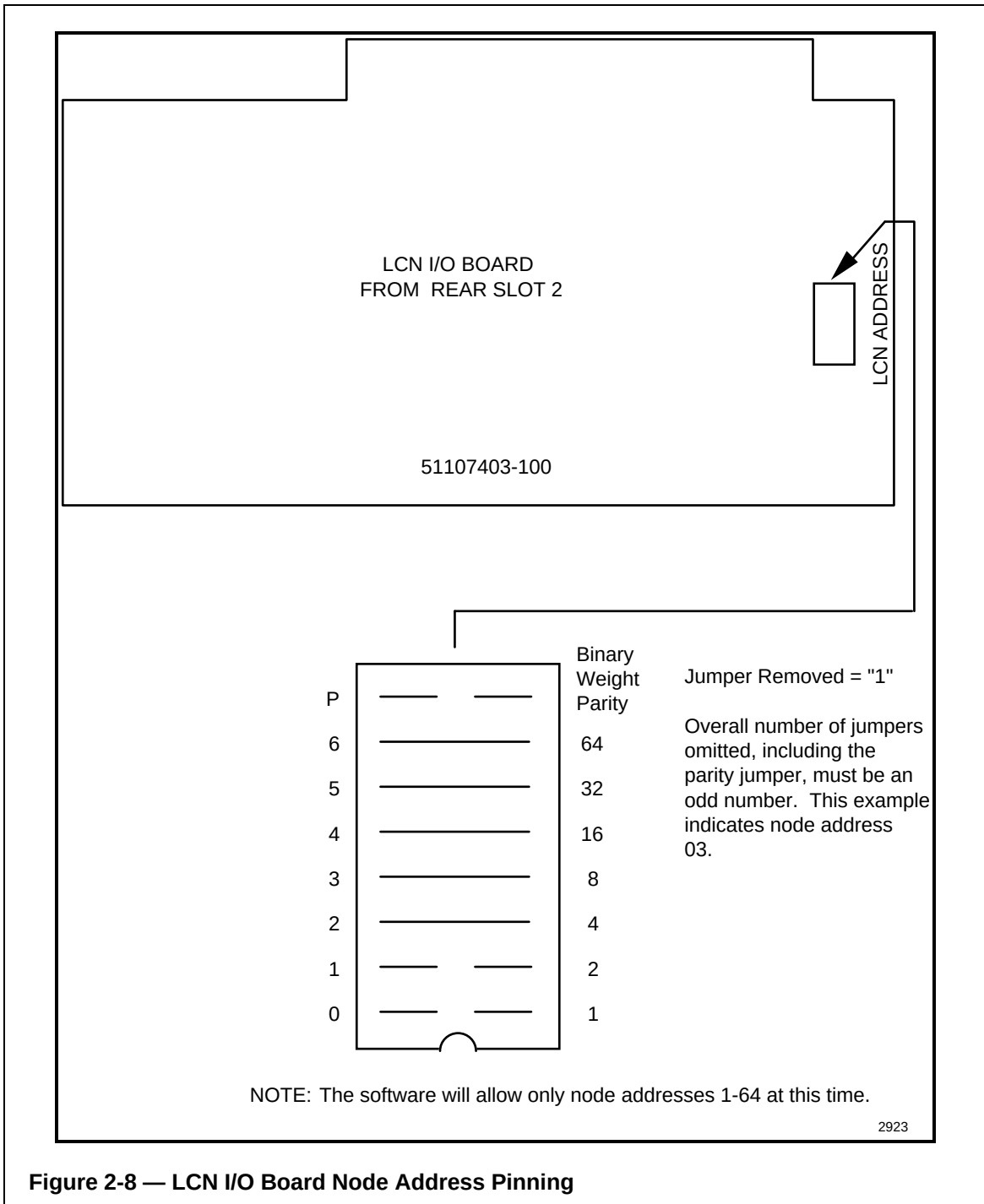
The address pinning is logically ANDed together if an LCN I/O board is used with a K2LCN or K4LCN board. **The address should be configured only on the LCN I/O board.**

For Data Hiway device addressing and response priorities, refer to *Data Hiway Subsystem Planning* manual. For the Process Interface Units (PIUs), refer to HLPIU-M, LLPIU-M, or LEPIU-M Service manuals.

For peripheral device pinning or function/selection, refer to the *Universal Station Service* manual.

For keyboard pinning options on the Universal Work Station, refer to Appendix A, paragraph A.7.

Figure 2-9 illustrates LCN node address selection on the LCN I/O board. The figure illustrates an LCN node address of three (3). The absence of a jumper is a binary “1.” Parity must be an odd number.



2.9 CLCN A/B BOARD NODE ADDRESS PINNING (CE Compliant)

The CLCNA/B board is the CE Compliant version of the LCN I/O board.

Figure 2-9 illustrates LCN node address selection on the CLCNA/B board. The figure illustrates an LCN node address of three (3). Placing a switch in the zero (0) position is a binary “1.” Parity must be an odd number.

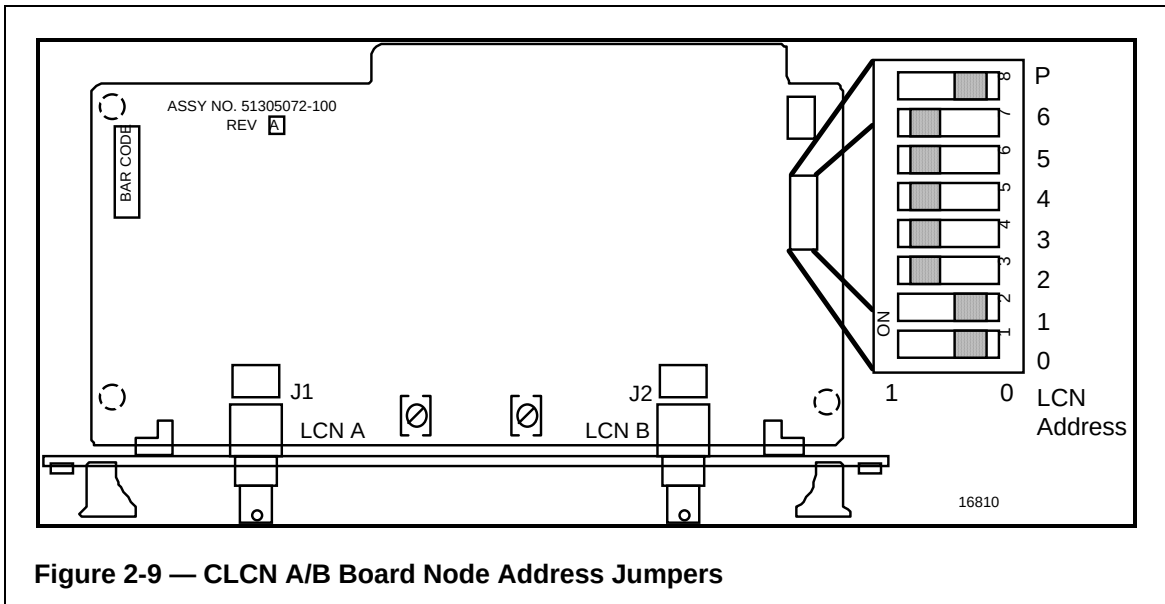


Figure 2-9 — CLCN A/B Board Node Address Jumpers

2.10 K2LCN/K4LCN BOARD NODE ADDRESS PINNING

If a node does not have an LCN I/O (or CLCNA/B) board, the LCN node address must be configured on the K2LCN or K4LCN board.

Figure 2-10 illustrates LCN node address selection on the K2LCN board, assembly 51401551-x00. The pinning block, TS2, is located at coordinate 9C on the board. The absence of a jumper is a binary “1.” Parity must be an odd number.

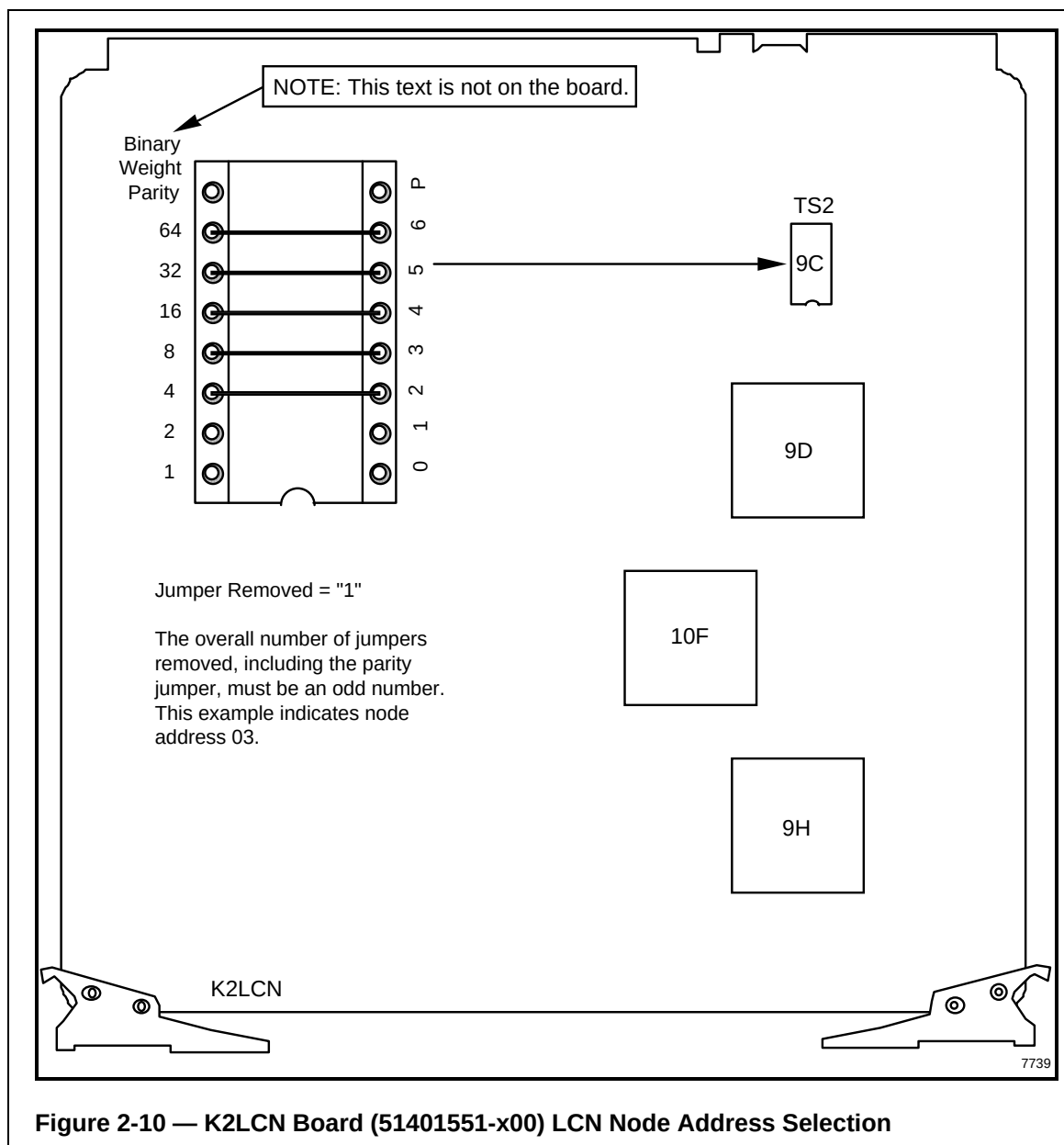


Figure 2-11 illustrates LCN node address selection on the K2LCN board, assembly 51402615-x00. The pinning block, TS1, is located at coordinate 7A on the board. The absence of a jumper is a binary “1.” Parity must be an odd number.

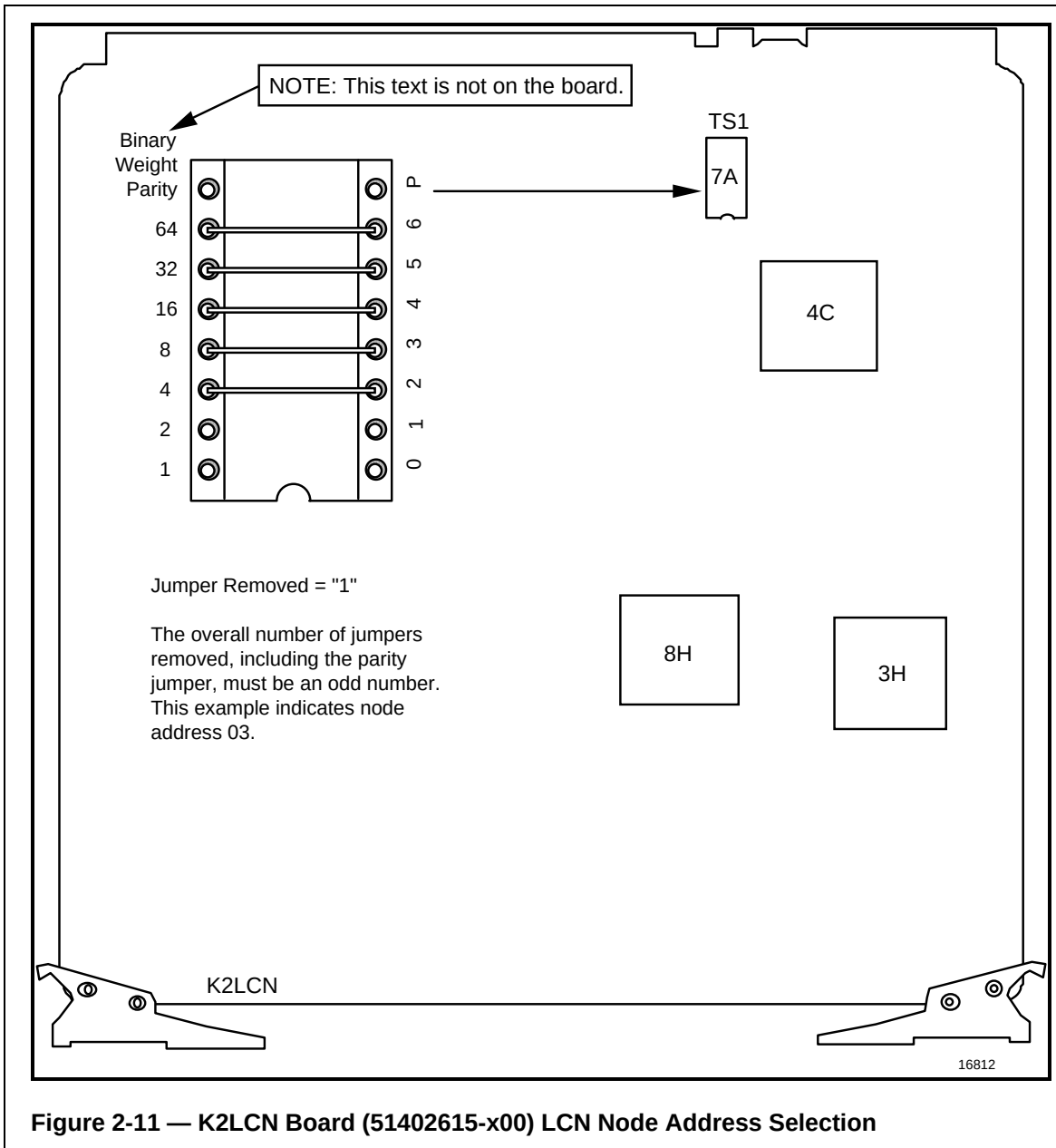


Figure 2-12 illustrates LCN node address selection on the K4LCN board, assembly 51401946-100. The pinning block, TS2, is located at coordinate 21E on the board. The absence of a jumper is a binary “1.” Parity must be an odd number.

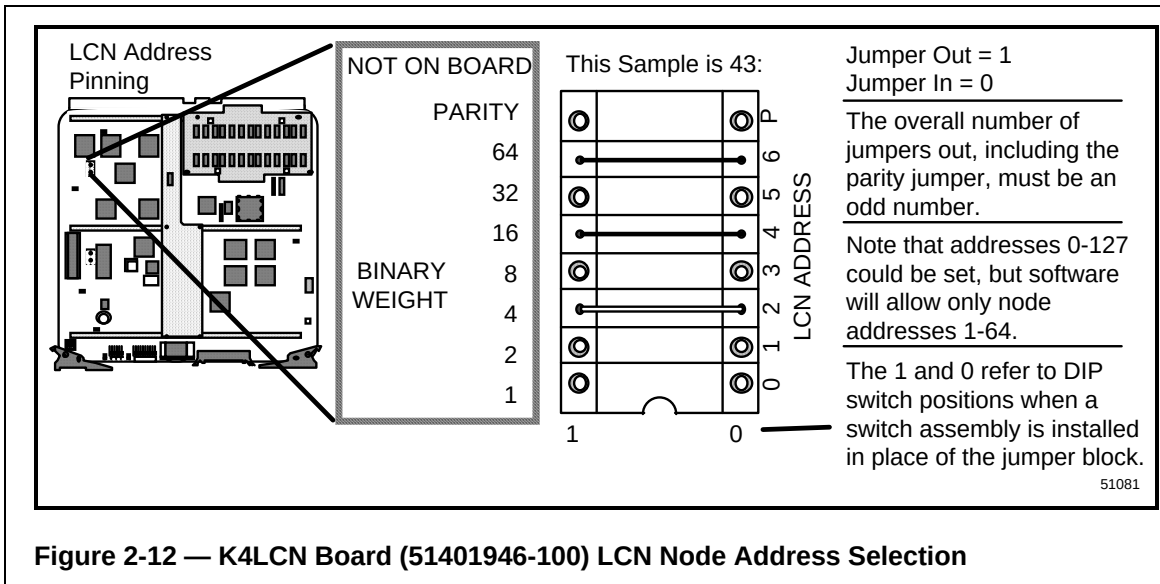
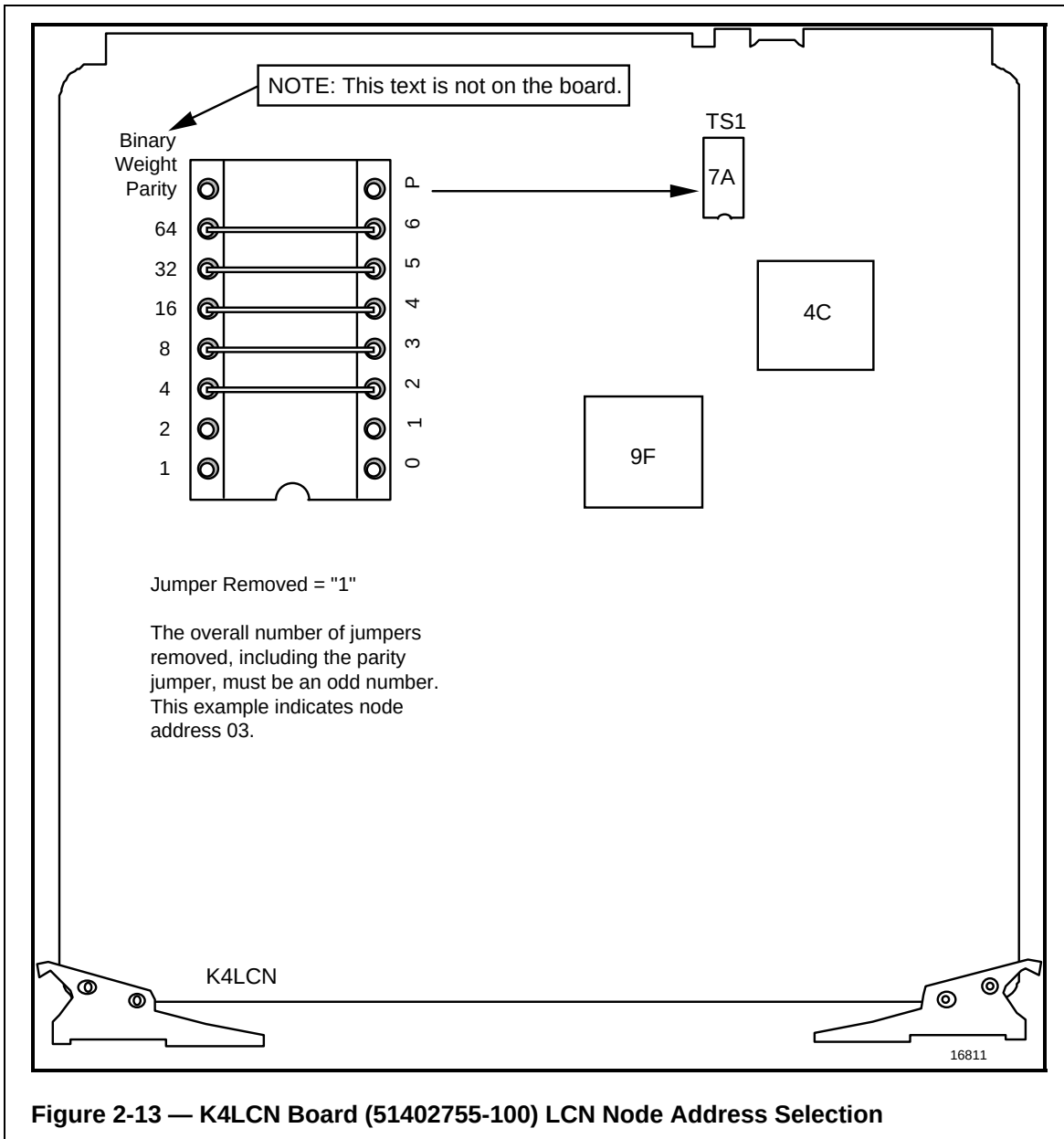


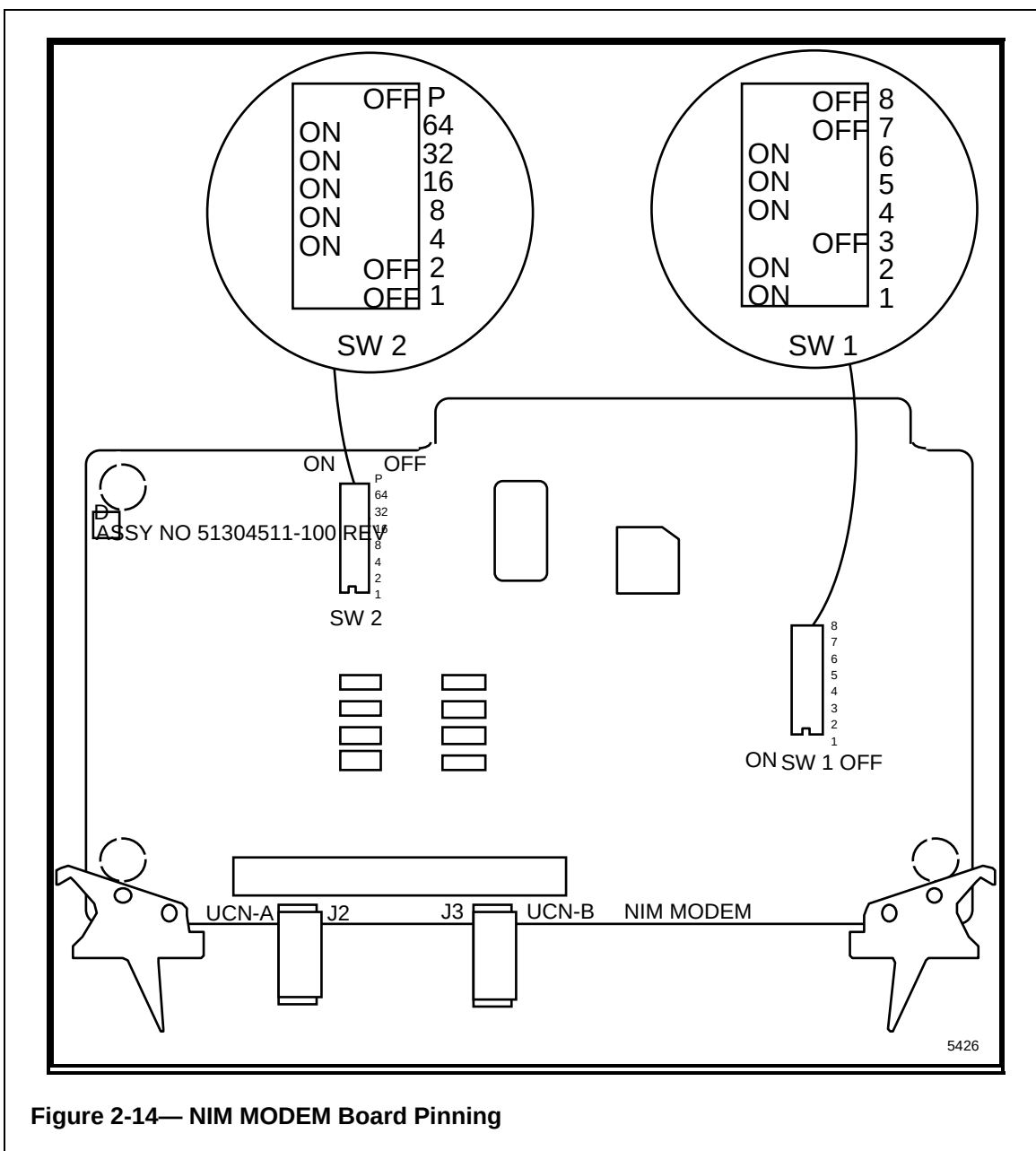
Figure 2-13 illustrates LCN node address selection on the K4LCN board, assembly 51402755-100. The pinning block, TS1, is located at coordinate 7A on the board. The absence of a jumper is a binary “1.” Parity must be an odd number.



2.11 NIM MODEM BOARD BOARD PINNING

Location SW1 has the revision pinning for the NIM Modem board. This pinning reflects the revision of the board and **is never to be changed unless the board is updated.**

Location SW2 has the pinning for the UCN node address. This address is a binary weighted number and is used only in the **Test** mode. The address entered, is the address of the primary NIM and must be an odd number with odd parity (an odd number of switches must be in the OFF position). See Figure 2-14.



2.12 NIM MODEM BOARD PINNING (CE Compliant)

The NIM board is the CE Compliant version of the NIM Modem board.

Location SW1 has the revision pinning for the NIM Modem board. This pinning reflects the revision of the board and **is never to be changed unless the board is updated.**

Location SW2 has the pinning for the UCN node address. This address is a binary weighted number and is used only in the **Test** mode. The address entered, is the address of the primary NIM and must be an odd number with odd parity (an odd number of switches must be in the OFF position). See Figure 2-15 for illustration of pinning.

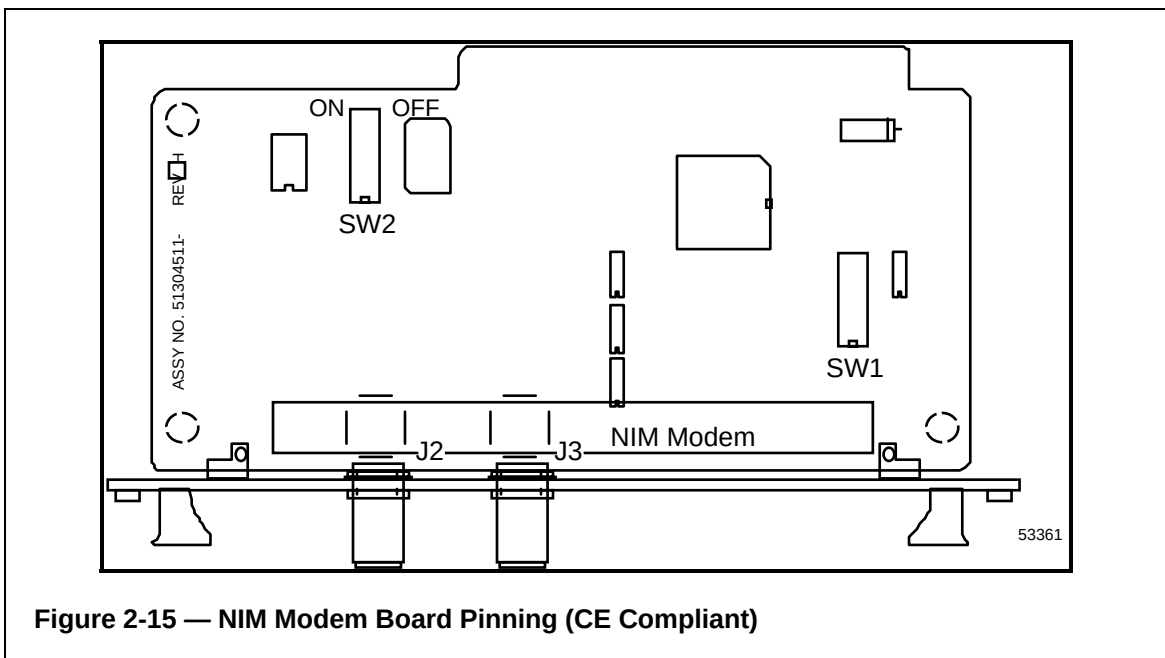


Figure 2-15 — NIM Modem Board Pinning (CE Compliant)

2.13 EPNI2 I/O BOARD PINNING

Location SW1 has the pinning for the EUCN node address. This address is a binary weighted number and is used only in the **Test** mode. Enter the EUCN node address of the primary ENIM. It must be an odd number with odd parity (an odd number of switches must be in the OFF position). See Figure 2-16.

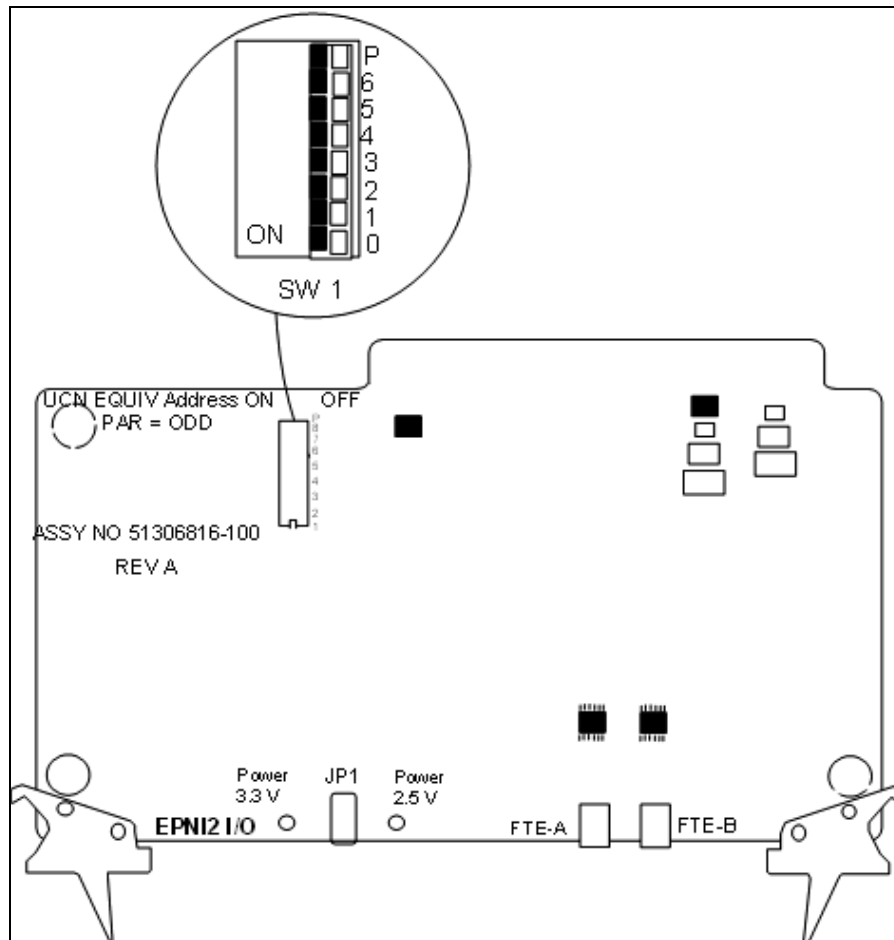


Figure 2-16 — EPNI2 I/O Board Pinning

The CE Compliant EPNI2 I/O board pinning is illustrated in Figure 2-17.

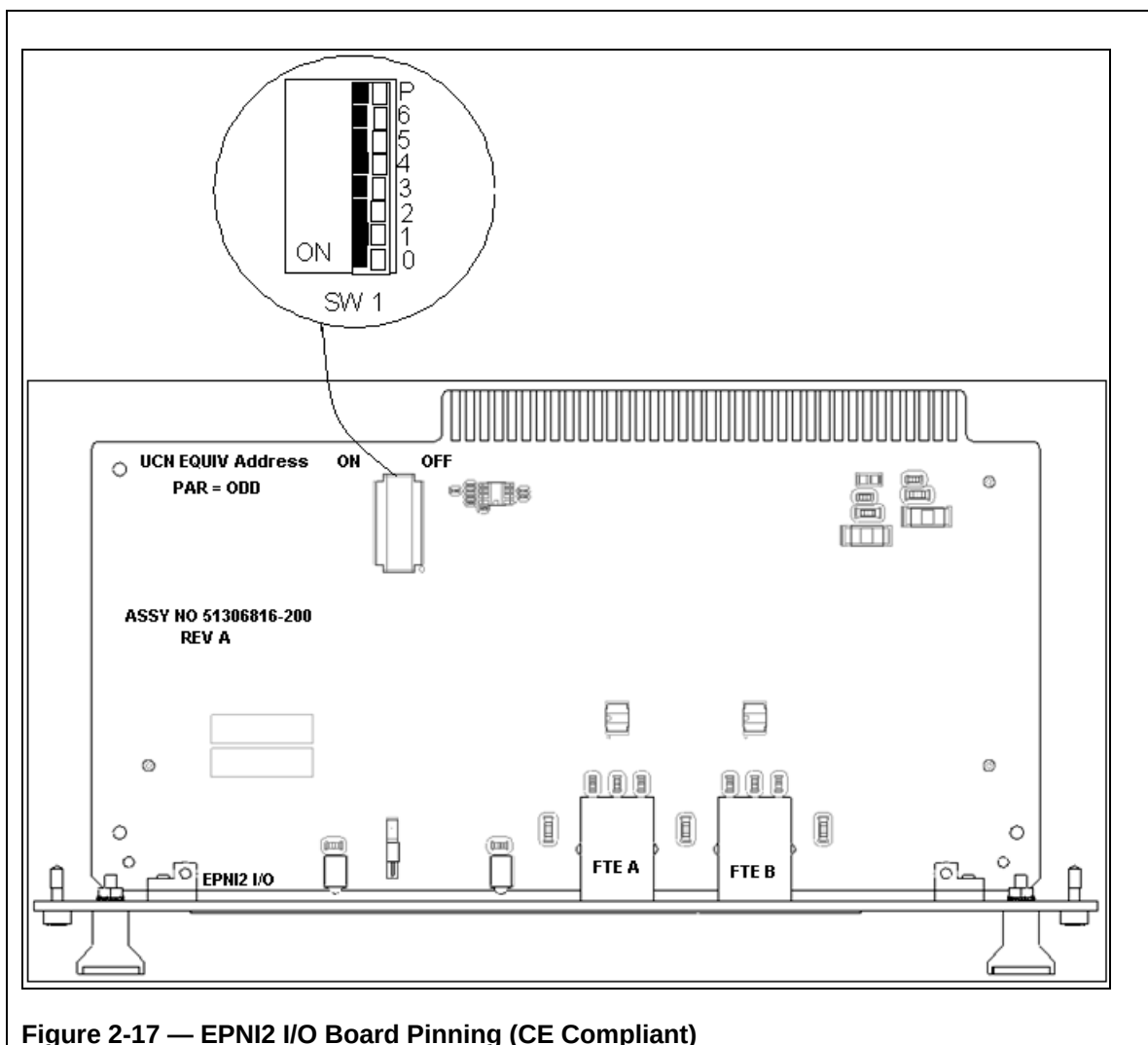


Figure 2-17 — EPNI2 I/O Board Pinning (CE Compliant)

2.14 WORK STATION INTERFACE (WSI) BOARD PREPARATION

The WSI board has a Normal Shutdown switch that should be in the NORMAL position. The SHUTDOWN position disables the Work Station Processor and returns the node to a UNP personality.

WSI boards have a real-time battery that must be activated as shown in Figure 2-18.

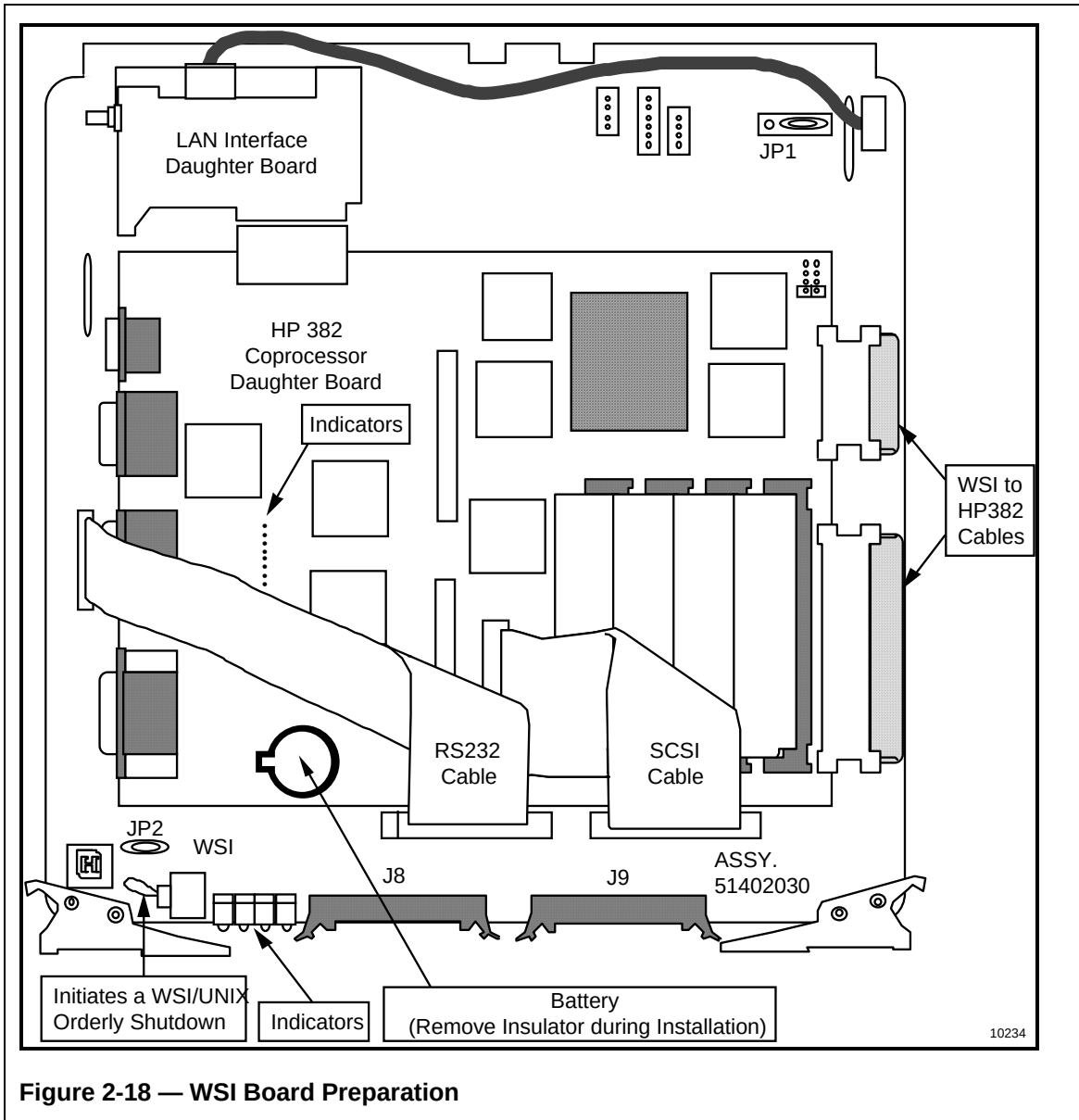


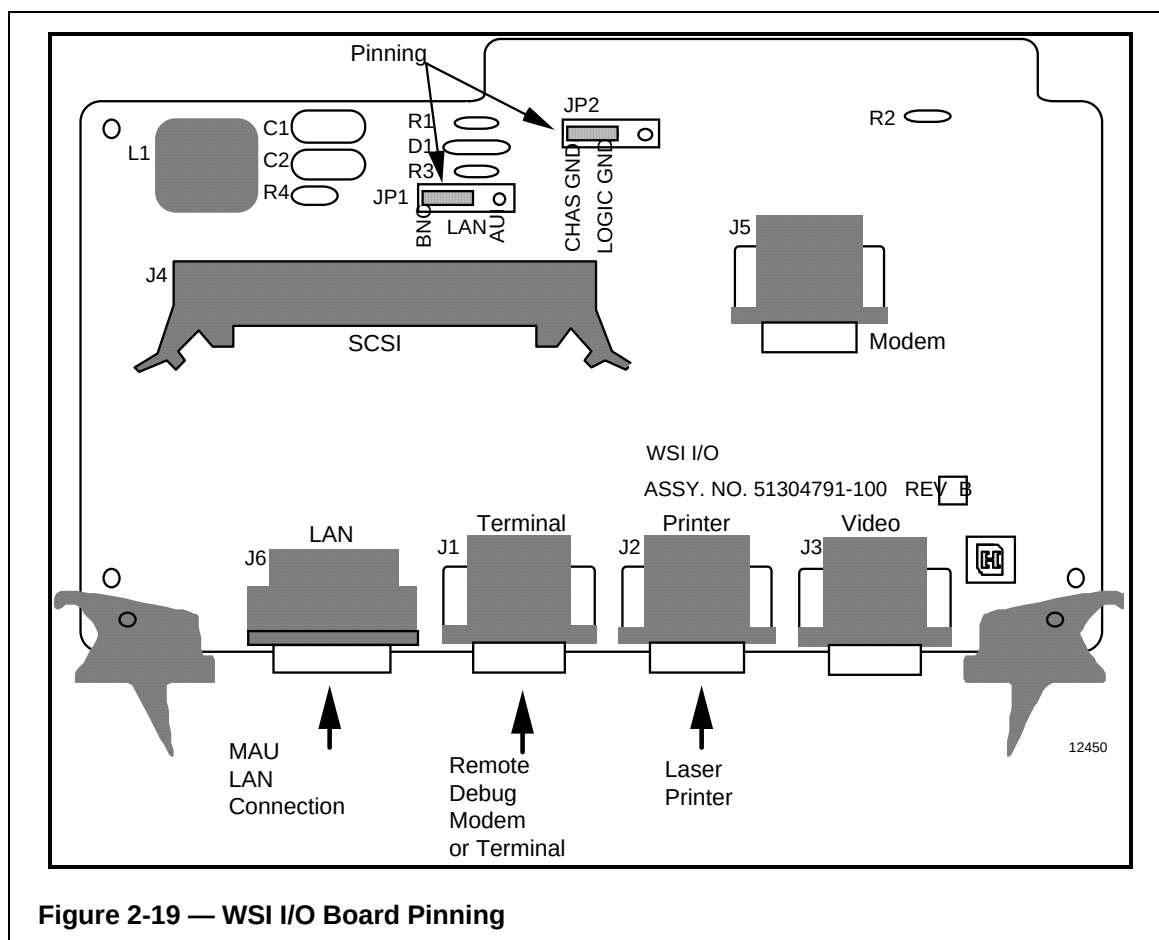
Figure 2-18 — WSI Board Preparation

2.15 WORK STATION INTERFACE I/O (WSI I/O) BOARD PINNING

A Universal Station^X node contains a WSI I/O board that is to be pinned as shown in Figure 2-17.

The WSI I/O has pinning required:

- for LAN connection type
 - BNC type cable connection
 - AUI type cable connection
- for Logic ground connection to chassis ground
 - In a Universal Station^X connect logic ground to chassis ground

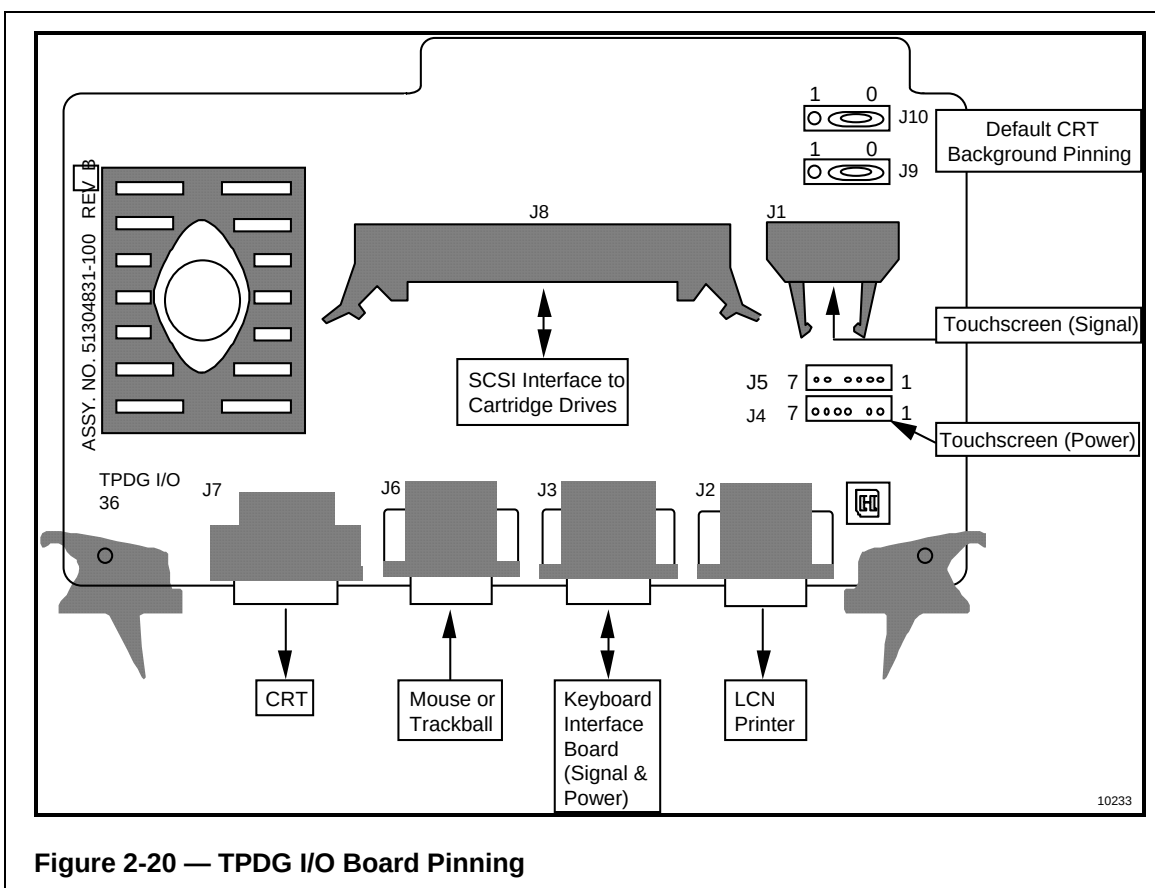


2.16 TPDG I/O BOARD PINNING

TPDG I/O pinning selects the monitor background color present before the color palate is activated. Four shades of gray are selected using J10 and J11 in combinations.

COLOR	J10	J11
Black	0	0
Light Gray	0	1
Medium Gray	1	0
Warm Gray	1	1

Figure 2-20 illustrates the pinning on the TPDG I/O board.



2.17 TPDG I/O BOARD PINNING (CE Compliant)

The TPDGC is the CE Compatible version of the TPDG I/O board.

TPDGC pinning selects the monitor background color present before the color palate is activated. Four shades of gray are selected using J7 and J8 in combinations.

Figure 2-21 illustrates the pinning on the TPDGC I/O board.

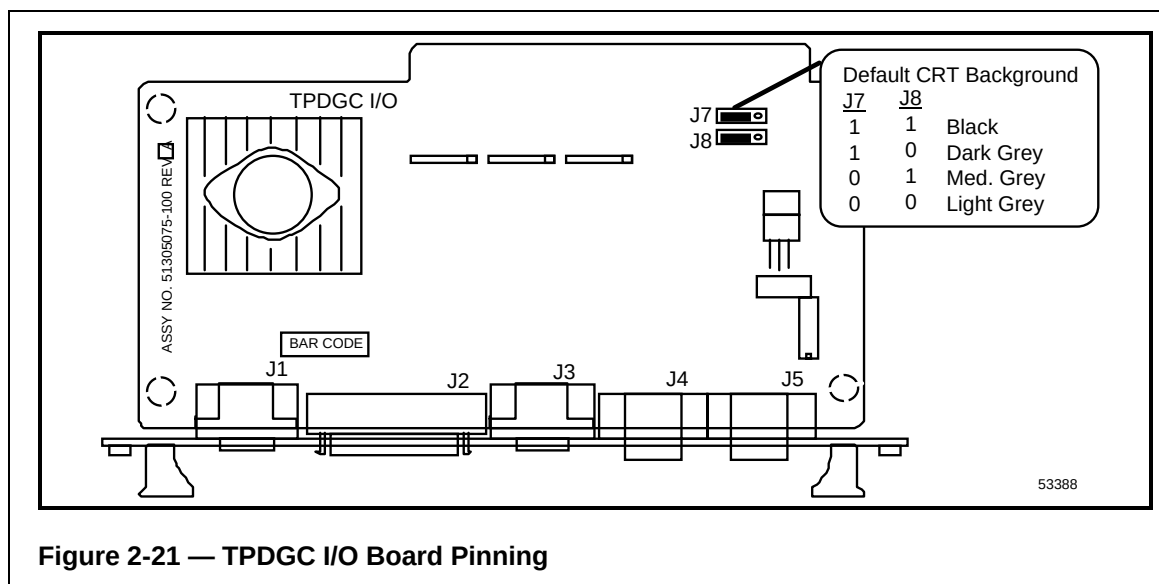


Figure 2-21 — TPDGC I/O Board Pinning

2.18 WSI2 BOARD PINNING

WSI2 has one jumper. It should be installed as shown below.

Processor Type	J10
HMPU	Out
K2LCN	In
K4LCN	In

Figure 2-22 illustrates the pinning on the WSI2 board.

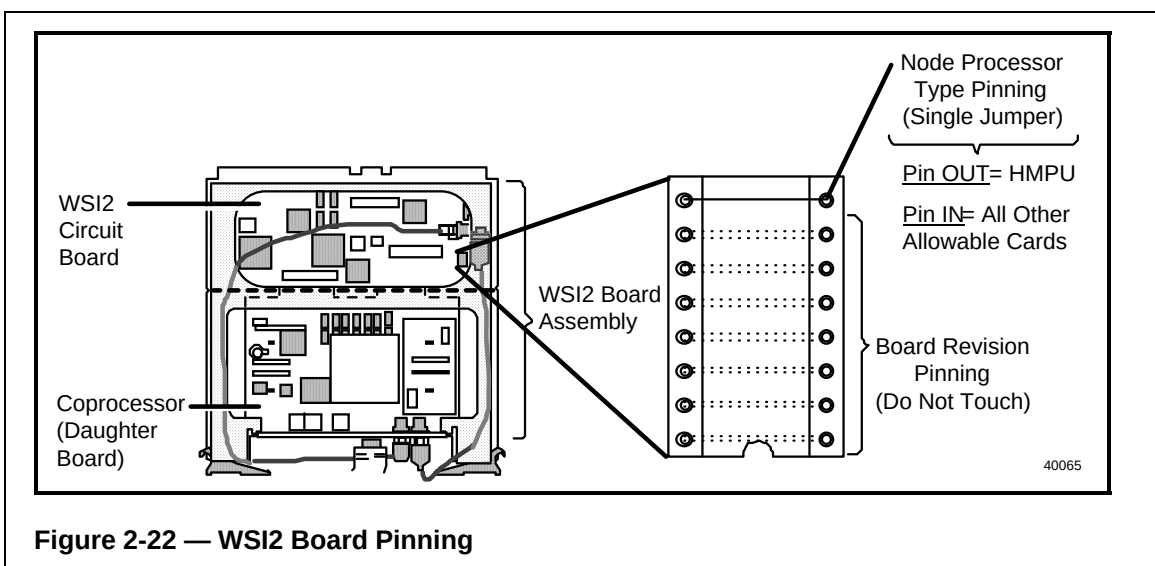


Figure 2-22 — WSI2 Board Pinning

2.19 PRECISION CLOCK PINNING (POWER SUPPLY REPLACEMENT)

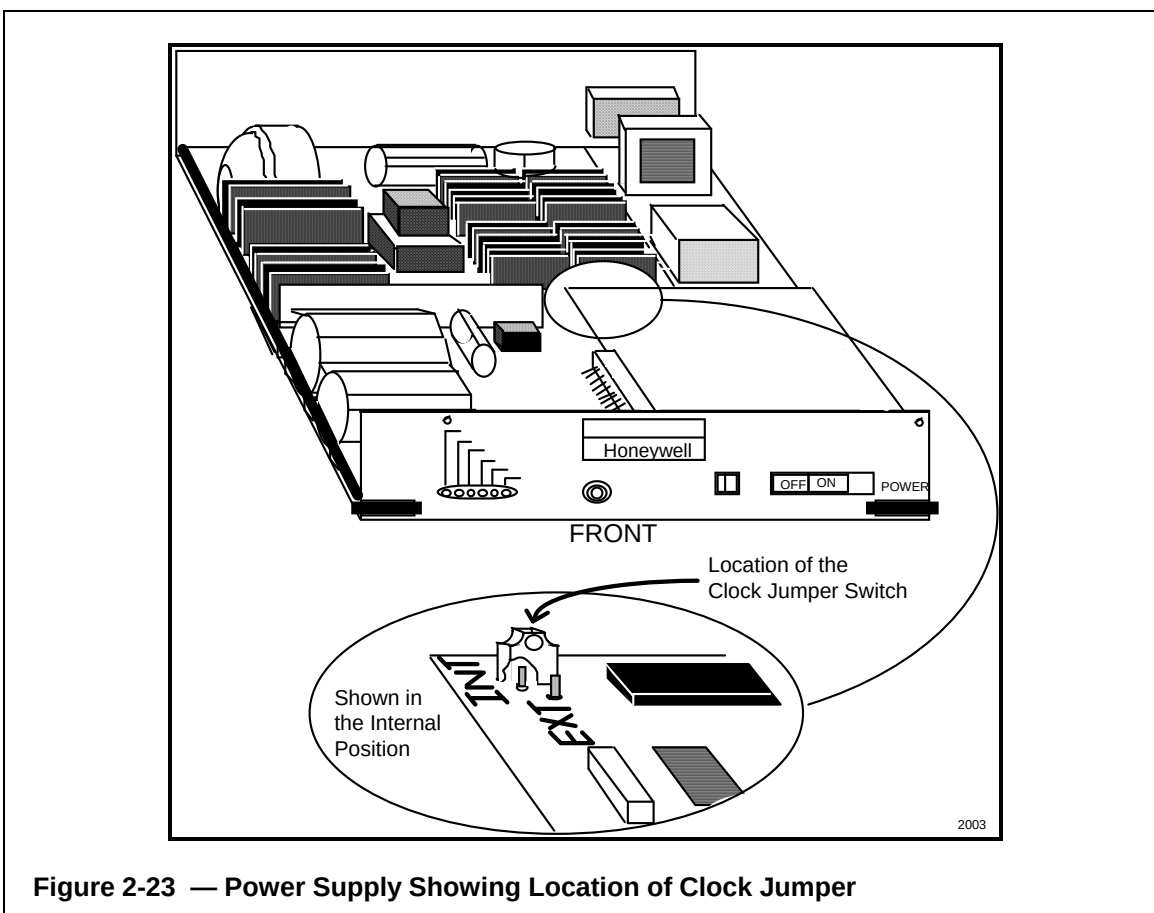
NOTE

New Five-Slot and Ten-Slot power supplies that are CE compliant went into production in 1995. These power supplies have precision clock pinning just as their predecessors.

The usual system configuration has the clock pinning in the power supply configured for an internal clock (see Figure 2-21, 2-22, 2-24, or 2-25).

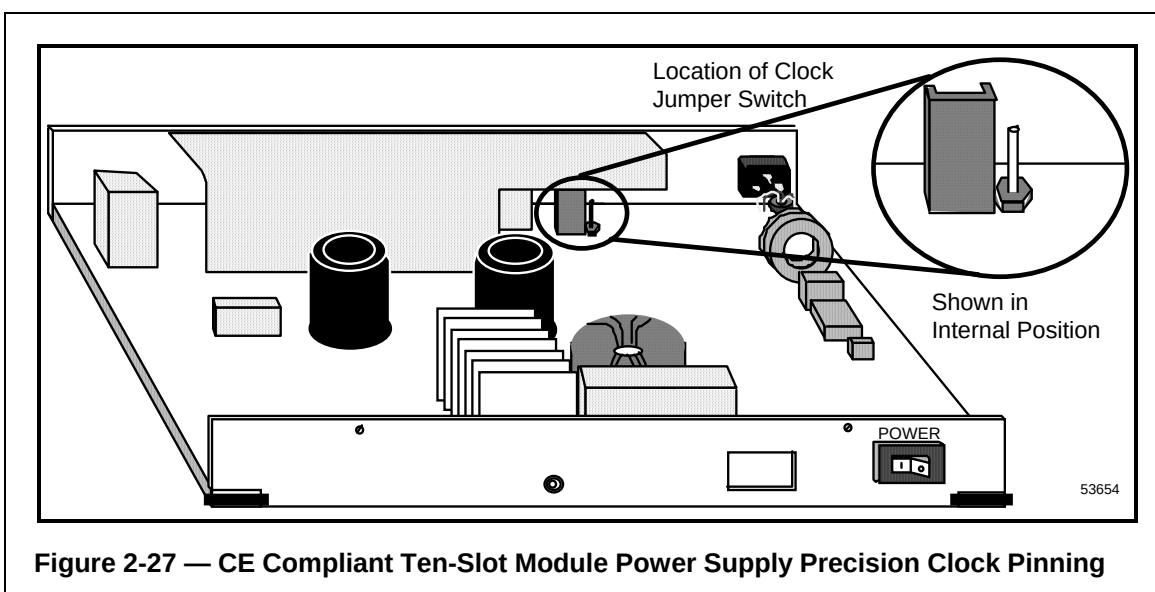
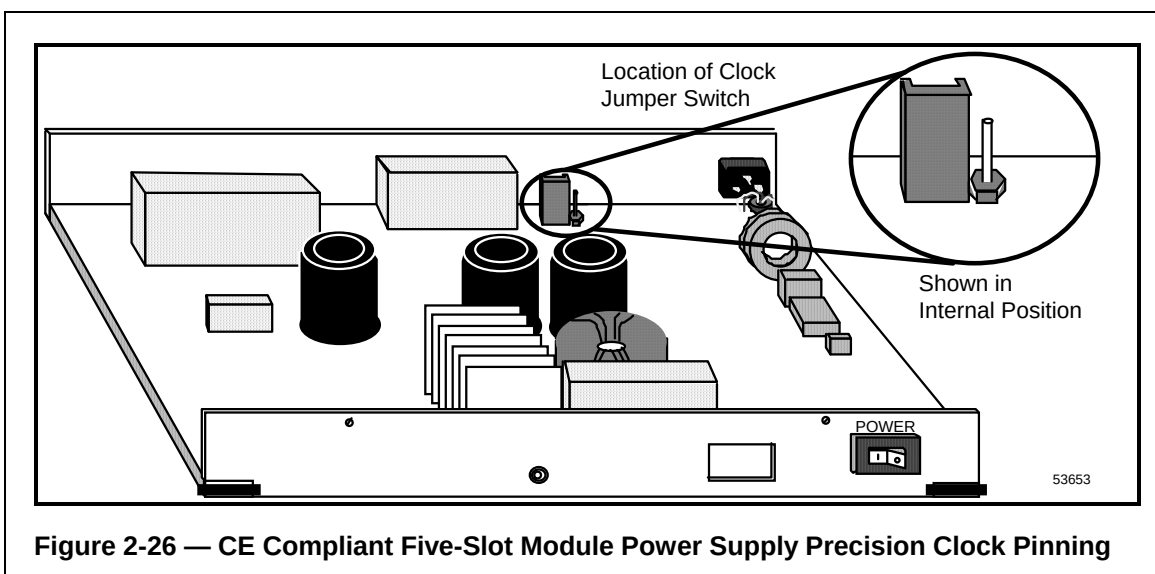
If the Precision Clock Source option is installed, this pinning is changed from internal to external. Also, when this option is installed, a label is placed on the back of the module listing the option. In addition, there is a cable connecting JC on the Precision Clock Source card (located in an I/O slot at the rear of the module) to connector J7 on the rear chassis of the module, as shown in Figure 2-23.

If you replace the power supply, verify the clock option you are using and check that either internal or external pinning has been selected as in Figure 2-21, 2-22, 2-24, or 2-25. Note you may be using one of four different power supply designs, but the pinning differences between them are easily recognized.

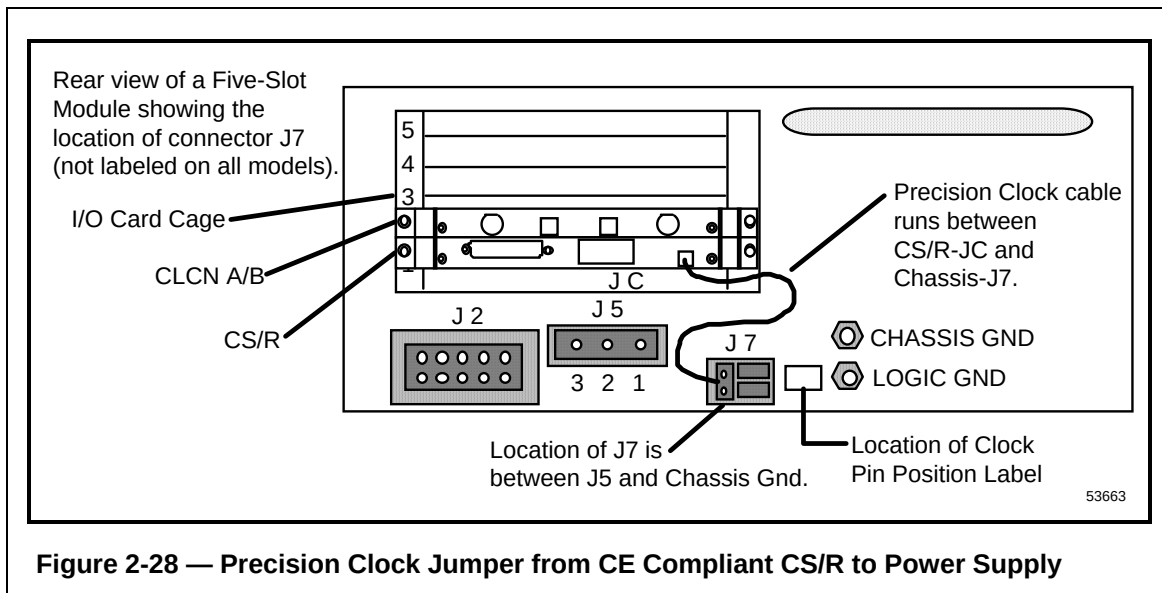


2.20 CE COMPLIANT POWER SUPPLY PRECISION CLOCK PINNING

The CE Compliant power supplies are pinned as shown in Figure 2-24 and 2-25. The cable from the CS/R is the same as with the older supplies.



Connect the cable between JC on the CS/R board and J7 on the card file as shown in Figure 2-28.



2.21 MODULE GROUNDING

Refer to *LCN System Installation* manual for grounding information for Five-Slot and Ten-Slot Modules.

TEST/TROUBLESHOOTING

Section 3

3.1 TESTS

The Module is tested through firmware (in hardware) and software checks. Firmware tests built into the module provide two similar means of functionally checking the unit, whether or not it is connected to the LCN. The first firmware test starts when power is applied. Pressing the **RESET** button initiates a second (slightly different) firmware test.

Software tests start after the module is connected to the system. Loading the module personality, for example, initiates a software Quality Logic Test (QLT).

Figure 3-1 is a flow diagram showing the relationship of the hardware and software tests.

3.2 TEST PROCEDURES

NOTE

Each module must have a processor, memory, LCN interface, and LCN input/output paddle-board installed before it can be tested. Some functions, such as processor and memory, may be physically located on the same board.

Initiate power-up tests by setting the power switch to on. Note that the red LED indicators on the boards light for a few seconds (less than 1.5 minutes). Then they should turn off and the green LED indicators turn on as the node passes its Quality Logic Tests. When the tests are complete, each node's alphanumeric display indicates its node address.

Check the power supply **ERROR** and **POWER OK** LED indicators and the **FAN ALARM** LED indicator to ensure that no red LED indicators are on and that the green **POWER OK** LED indicator lights.

If this is not a History Module, press the **RESET*** button. The red LED indicators momentarily go on as above, then go out when the green LED indicators come on. Recheck the power supply, board, and fan LED indicators as above. Note that each alphanumeric display indicates its node address.

If any of the above conditions does not pass, refer to subsection 3.3, *Troubleshooting*.

* **WARNING:** In the case of the History Module only, if you press the RESET button indiscriminately while the system is operating, you may cause damage to the data stored on the hard disk. See subsection 3.2, *Troubleshooting* in the *History Module Service* manual for complete details.

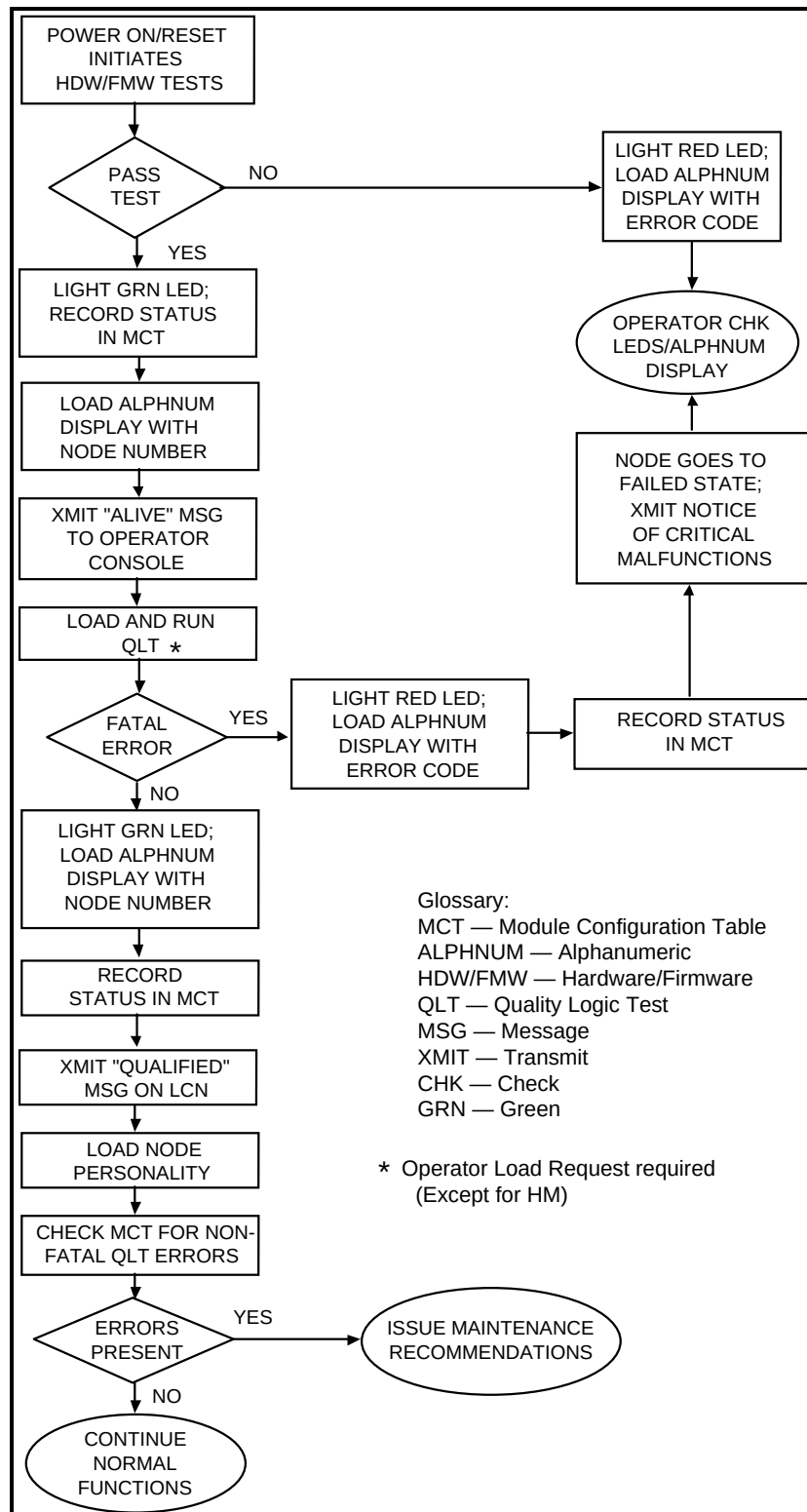


Figure 3-1 — Testing Flow Diagram

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3.3 TROUBLESHOOTING

The troubleshooting procedures are grouped by affected ORU and are keyed to the LED indicators and the alphanumeric display. First, a malfunction symptom is listed, then instructions are given to check or replace parts in order of the most likely causes. In the following procedures, the indicated LED indicators are red unless otherwise stated. Refer to subsection 4 for disassembly and assembly instructions. Always turn power off before removing or replacing a part on the module, as this protects equipment from voltage transients and ensures proper initialization. If you can't correct the fault by performing these procedures, refer to Appendix A for alphanumeric display troubleshooting. If you still cannot isolate the fault, contact the Honeywell Technical Assistance Center (TAC).

CAUTION

All assemblies used in this Module are likely to contain electrostatically sensitive devices. Use a personnel grounding strap and grounded work surfaces and equipment. Store and transport parts only in electrostatically safe containers.

HMPU, HPK2-2, HPK2-3 boards have LED indicators at the front of the board to indicate error conditions. The boards also have a diagnostic bar display indicators with no labels as shown and defined in Figure 3-2.

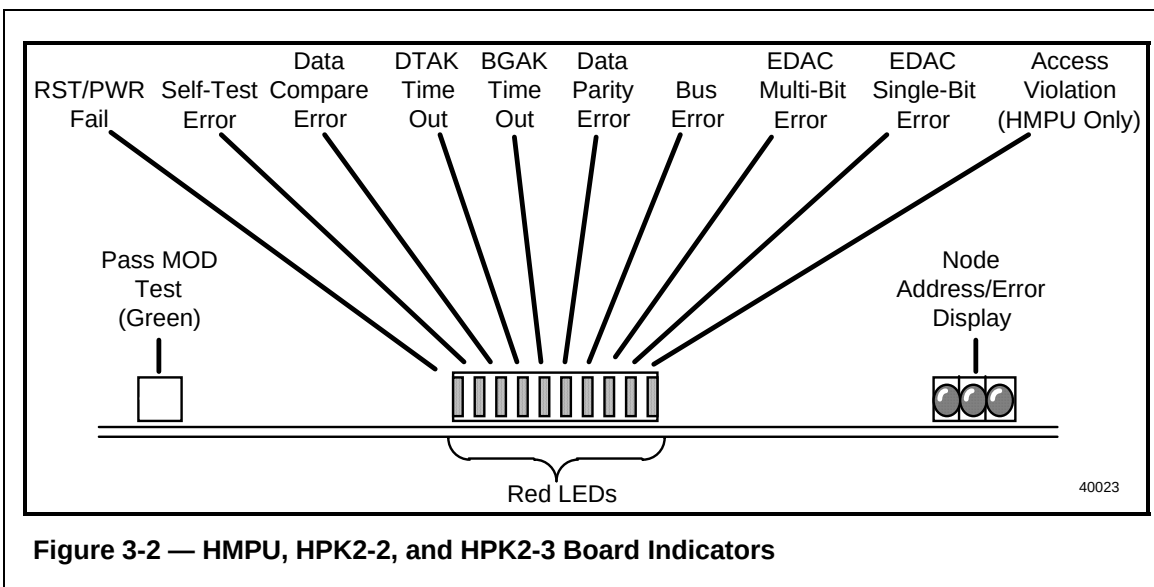


Figure 3-2 — HMPU, HPK2-2, and HPK2-3 Board Indicators

The K2LCN and K4LCN boards have the same LEDs and diagnostic bar display indicators with one exception. The “Access Violation” indicator is replaced by the “LCN Transaction Error” indicator at the far right of the diagnostic bar display. Figure 3-3 defines the indicators for early production K2LCN and K4LCN boards, assemblies 51401551-x00 and 51401946-100, respectively.

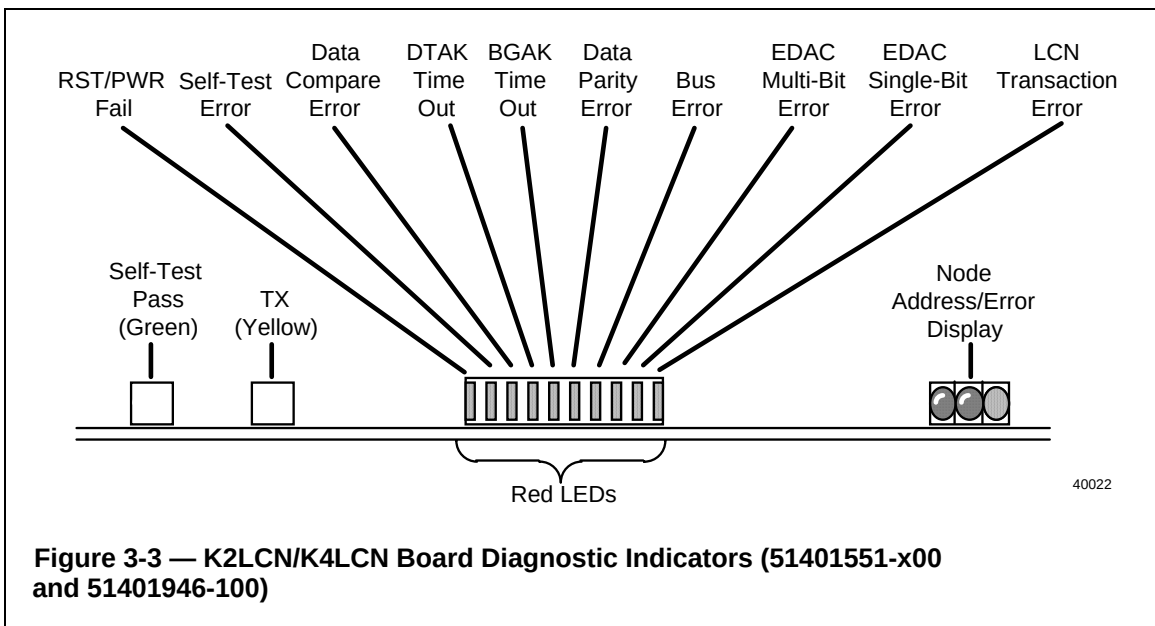


Figure 3-4 defines the indicators for later production K2LCN and K4LCN boards, assemblies 51402615-x00 and 51402755-100, respectively.

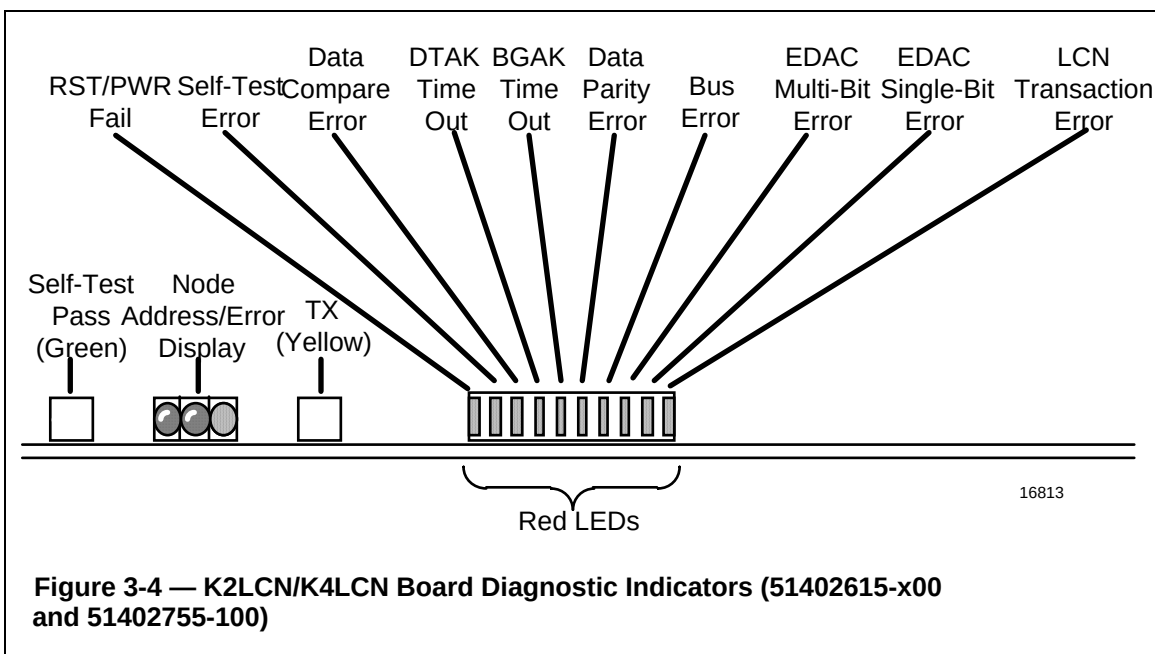


Figure 3-5 defines the LED and diagnostic bar display indicators on the AMR board that is used in the Application Module.

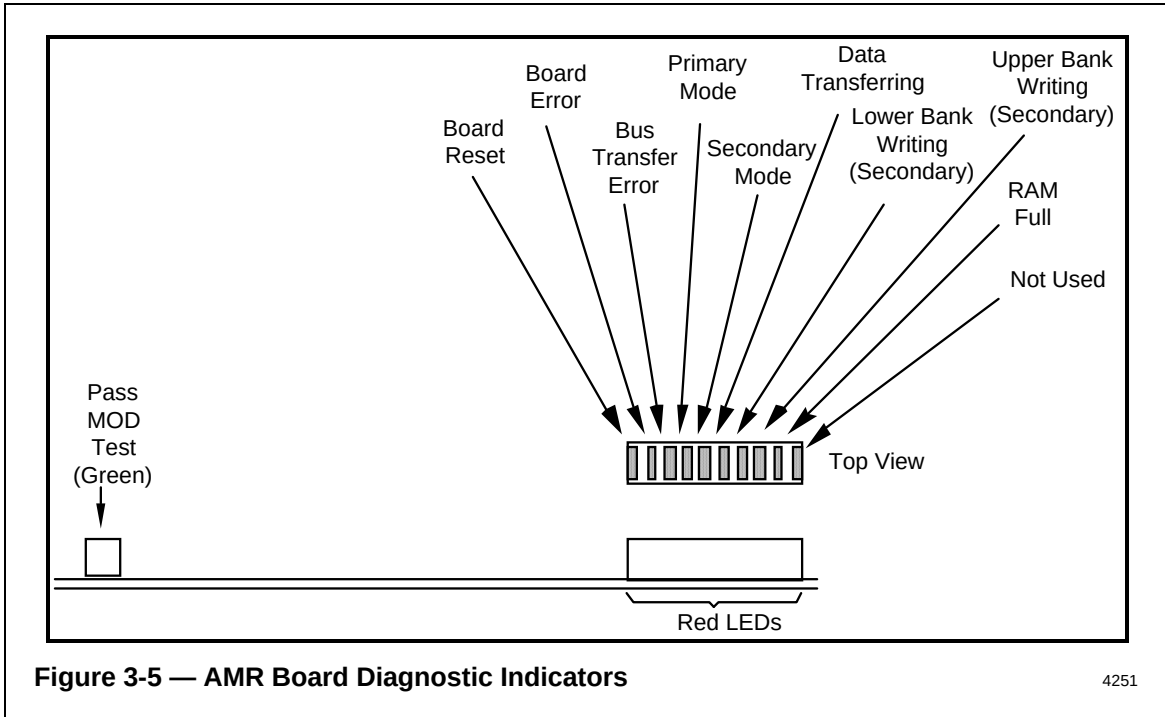
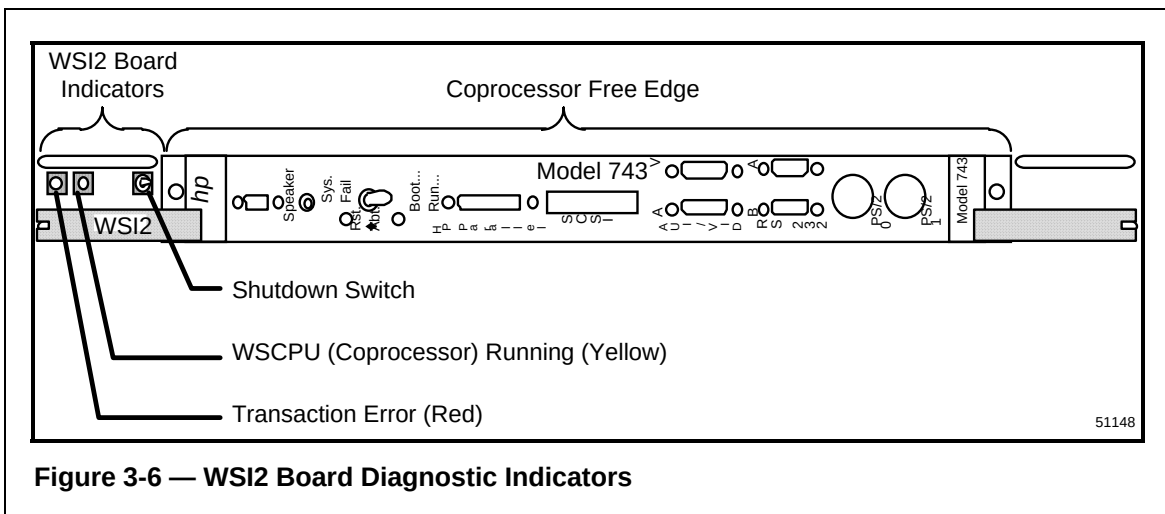


Figure 3-6 defines the LED and diagnostic bar display indicators on the WSI2 board that is used in the Universal Station^X.



3.3.1 Power Supply/Fan Faults

No LEDs are lit.

- Check the power cord and power switch.
- Replace the power supply.

Any power-status panel LED (but not **OVER TEMP**) on or green **PWR OK** LED is off.

- Replace the power supply.

The **OVER TEMP** or **FAN ALARM** LED is lit

WARNING

Keep fingers and other objects clear of fans. They have rotating parts and sharp fan blades.

- With power applied to the module, loosen the two quarter-turn bail-headed fasteners that are holding the fan assembly and quickly slide the assembly out of the module. If the fans are not binding, the fan blades will continue turning for a few seconds. After the fans have stopped, check for bearing drag by attempting to rotate the blades with your fingers.
- If one or more fans are not turning or if bearings are dragging, replace the fan assembly.
- If problem is still not corrected, check the fan power connector at the backplane or replace the power supply.

NOTE

Remove the module front panel to view the board LEDs. Check the alphanumeric display. The first two digits are the failed-unit slot number and the last digit is the failed test number.

3.3.2 Controller Board (LCN, FDC, WDC, DHI, CLI, CNI, SIO, SPC) Faults

SELF TST/ERR indicator does not go out.

- Ensure that the I/O board is properly installed.
- Replace the controller board.
- Replace the I/O board. Note: The LCN I/O board has a 14-pin bus-wire node address DIP on the right side. When replacing an LCN I/O board, transfer the old DIP or individual clips to the new replacement board.

SELF TST/ERR indicator goes out, but the **PASS MOD TEST** (green) indicator does not light on a controller board.

- Replace the controller board and boards in the slots below it one at a time.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

SELF TST/ERR indicator goes out, but the **PASS MOD TEST** (green) indicator does not light on all controller boards.

- Replace the HMPU, HPK2, K2LCN, or K4LCN board.
- Remove the top controller board, then apply power. Repeat for each controller board until the fault is cleared. Retry the extracted boards.

BUS TRAN ERR indicator is lit on the controller board and the **DTAK TIME OUT** indicator is lit on the HMPU, HPK2, K2LCN, or K4LCN board.

- Replace controller board and the boards in slots below it one at a time.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

BUS TRAN ERR indicator is lit on a the controller board and the **BGAK TIME OUT** indicator is lit on the HMPU, HPK2, K2LCN, or K4LCN board.

- Check for more than two open slots between the boards.
- Remove the top controller board, then apply power. Repeat for each controller board until the fault is cleared. Retry extracted the boards.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

3.3.3 PIC and VDG Board Faults

SELF TST/ERR indicator does not go out on the PIC or VDG board.

- Replace the PIC board.
- Replace the VDG board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

SELF TST/ERR indicator goes out on the PIC board and the **PASS MOD TEST** (green) indicator is on, but the **SELF TST/ERR** indicator is lit on the VDG board.

- Ensure that the VDG-PIC cables are in place and tight on the front of the unit.
- Replace the VDG board.
- Replace the PIC board.

SELF TST/ERR indicator goes out on the PIC and VDG boards, but the **PASS MOD TEST** (green) indicator does not light on the PIC board.

- Replace the PIC board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

BUS TRAN ERR indicator is lit on the PIC board and the **DTAK TIME OUT** indicator is lit on EMPU, HMPU, HPK2, K2LCN, or K4LCN board but the **BUS ERR** indicator is not on.

- Replace the PIC board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board

BUS TRAN ERR indicator is lit on the PIC board and the **DTAK TIME OUT** and **BUS ERR** indicators are lit on the EMPU, HMPU, HPK2, K2LCN, or K4LCN board.

- Replace the PIC board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

3.3.4 Memory Board (MMEM, EMEM, PMEM, QMEM) Faults

Refer to the configuration table (subsections 2.2.2 through 2.2.27) for your module and verify the memory boards are in the correct slots. Also check each memory board for compatibility with other memory boards. Refer to Table 2-5.

PASS MOD TEST indicator (green) does not light on one board.

- See if the board operates in another chassis slot.
- Replace the MMEM, EMEM, PMEM, or QMEM board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

MULT BIT ERR indicator is lit on the MMEM, EMEM, PMEM, or QMEM board and the **DTAK TIME OUT** and **BUS ERR** indicators are lit on the EMPU, HMPU, or HPK2 board, or just the **MULT BIT ERR** and **BUS ERR** indicators are lit.

- Replace the MMEM, EMEM, PMEM, or QMEM board.
- Replace the HMPU, HPK2, K2LCN or K4LCN board.

SING BIT ERR indicator is lit.

- Replace the MMEM, EMEM, PMEM, or QMEM board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

MULT BIT ERR indicator is lit on the MMEM, EMEM, PMEM, or QMEM board and the **DTAK TIME OUT** indicator is lit on the EMPU, HMPU, HPK2, K2LCN, or K4LCN board, but the **BUS ERR** indicator is not lit.

- Replace the MMEM, EMEM, PMEM, or QMEM board.
- Check to see if a **BUS TRAN ERR** indicator on a controller board is lit and replace the board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

3.3.5 Processor Board (EMPU, HMPU, HPK2, K2LCN, or K4LCN) Faults

DATA PAR ERR and **BUS ERR** indicators are lit on EMPU, HMPU, HPK2, K2LCN, or K4LCN board.

- Check the alphanumeric display. The first two characters denote the slot number of the failed board. The third character is the failed test number. Refer to Appendix A for the specific alphanumeric codes.
- Replace the indicated board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

3.3.6 PDG , EPDG, TPDG, and TPDGX Board Faults

SELF TST/ERR indicator does not go out on the PDG/EPDG board.

- Replace the I/O card (EPDG I/O, EPDGP I/O, TPDG I/O, or TPDGC).
- Replace the PDG, EPDG, TPDG, or TPDGX board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

SELF TST/ERR indicator goes out on the PDG, EPDG, TPDG, or TPDGX board, but the **PASS MOD TEST** (green) indicator does not light on the PDG, EPDG, TPDG, or TPDGX board.

- Replace the I/O card.
- Replace the PDG, EPDG, TPDG, or TPDGX board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

BUS TRAN ERR indicator is lit on the PDG, EPDG, TPDG, or TPDGX board and the **DTAK TIME OUT** indicator is lit on the HMPU, HPK2, K2LCN, or K4LCN board, but the **BUS ERR** indicator is not lit.

- Replace the I/O card.
- Replace the PDG, EPDG, TPDG, or TPDGX board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

BUS TRAN ERR indicator is lit on the PDG, EPDG, TPDG, or TPDGX board and the **DTAK TIME OUT** and **BUS ERR** indicators are lit on the EMPU, HMPU, HPK2, K2LCN, or K4LCN board.

- Replace the I/O card.
- Replace the PDG, EPDG, TPDG, or TPDGX board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

3.3.7 EPNI and PNM Board Faults

The EPNI and PNM boards are controller boards which interface the bus and processor in a way that is similar to the controller boards listed in Section 3.3.2. First, check the **SELF TST/ERR** indicator (red; should be out) and the **PASS MOD TEST** indicator (green; should be on) on the EPNI and PNM boards.

SELF TST/ERR indicator (red) is driven by a microprocessor on the EPNI board. If it is lit, check for the following causes:

- There was a hardware failure on the EPNI board.
- A problem was detected on-line (for instance, there may have been a EPNI local RAM parity error or a duplicate address may have been detected).
- The node was shut down (stunned) due to a watchdog timeout.
- The number of raw errors detected exceeded a pre-set threshold, causing the software on the EPNI board to enter the failed state.

If the state of the **SELF TST/ERR** indicator and **PASS MOD TEST** indicator are correct, continue with this instruction. Under normal system operating conditions, the following indicators and connections are present on the EPNI or PNM boards.

- **Red LEDs** are out.
- **Green LED** is lit.
- **Yellow LEDs** either blink on and off (indicating traffic) or stay on (heavy traffic).
- **The ribbon cable** between the PNM and PNI I/O paddle boards is firmly fastened in place.
- **The two mini-coax cables** between the PNM board and the PNM I/O paddle board are firmly fastened in place.
- **The TX yellow LEDs** blink (or remain steadily on) during data traffic. The two indicators on the EPNI and PNM boards monitor identical circuits and blink or light in unison. Transmit data is sent simultaneously on both cables.
- **One of the RCVE CABLE yellow LEDs** on the PNM board blinks (or remains steadily on) as data traffic is received. The UCN signal is first received on one cable for about 5 minutes, then the receiver is switched to the other cable to maintain confidence.

Check that there are no disconnected or broken cables. If part of the UCN has failed, the failure reporting and diagnostic tests that is included in the software will help isolate the problem.

NOTE

ENIM differs from the NIM since it uses FTE cables that provide an IP Address. The ENIM node must receive an IP address to function properly. If there are no FTE cables connected, then the ENIM cannot be loaded.

Because of this difference between NIM and ENIM, the indication given by the left-most red LED on the front edge of the EPNI versus the EPNI2 board also differs. In the EPNI (NIM), if there are no coaxial UCN cables connected, the red LED light glows. In the EPNI2 (ENIM), the left-most red LED does not glow if there are no FTE cables connected. At least one FTE cable must be connected for the ENIM to receive an IP address.

3.3.8 EPNI2 Board Faults

The EPNI2 board performs a self-test. The self-test includes diagnostics to verify the operation of the I/O board, microprocessor kernel, and the support circuitry. In addition, it performs a loopback test of the data paths. The LEDs at the front of the EPNI2 board indicate the status.

Under normal circumstance the LEDs on the EPNI2 board displays the following:

LEDs	Status
SELF TST/ERR	Red is out
PASS MOD TEST	Green is on

SELF TST/ERR indicator (red) is driven by a microprocessor on the EPNI2 board. If the LED (red) is lit, check for the following causes:

- A hardware failure on the EPNI2 board during or after self-test.
- Any online problem (for instance, there may have been an EPNI2 local RAM parity error or duplicate address detection).
- The Node shut down (stunned) due to a watchdog timeout.
- The number of raw errors detected exceeded a pre-set threshold, causing the software on the EPNI2 board to enter the failed state.

Ensure that there are no disconnected or broken cables. If part of the EUCN fails, the failure reporting and diagnostic tests included in the software helps you to isolate the problem.

3.3.9 Fiber Optic Extender Board (XLCNE2, FOC/XMTR, FOC/RCVR CS/R) Faults

NOTE

The XLCNE2 board is a replacement for the LCNE2 board. The two boards perform the same function although they use different connectors. The allowable fiber optic cable lengths allowed are also different for each board. An XLCNE2 can only talk to another XLCNE2, and the LCNE2 and XLCNE2 cannot be mixed on the same fiber.

3.3.9.1 Normal Indicators

Under normal system operating conditions, the following indicators and switch settings are present on the boards that are a part of the LCN Extender Set.

- Red LEDs are out.
- Green LEDs are lit.
- Yellow LEDs are either blinking on and off (indicating traffic) or stay on (heavy traffic).
- RUN/RESET switch on the LCNE is in the run position.

3.3.9.2 Troubleshooting Indicators

If a remote segment has gone dead, check the following:

1. **Red LED on all LCNE boards should be off.** If a red LED is on, the Antijabber Flag is set. The failure that caused this to happen is on the board with the LED lit. Replace that board by using the procedures in *LCN Guidelines* manual. See Section 7, *LCNE and Fiber Optic Link Service*, in the manual.
2. **Green LED on LCNE, FOCR boards must be on steady.** If a green LED is not lit (or is flickering), there is not enough light power being received from the fiber optic cable. Check the following:
 - a. Cable integrity: Check for cut, damaged, or kinked cable.
 - b. Light received: Measure the light power at the receiving end of the cable. A minimum of 2.5 μ W must be measured.
 - c. Light transmitted: Measure the light power at the transmitter, a minimum of 50 μ W must be measured by using a one-meter length of fiber optic cable.

NOTE

If the Green LED is on continuously, there may still not be enough light power being received from the fiber optic cable. The Green LED is not an indication of signal quality. The Green LED indicates the presence of an attached cable.

3. Yellow indicators blink as data traffic is passed.

If a fiber optic connection that had previously worked ceases to do so, and no fault can be found with the LCNE/FOCT/FOCR boards, make the power measurements discussed in the *LCN Guidelines* manual. See Section 5, *Post-Installation Power Measurement*, in the manual.

3.3.10 NGI Board Faults

SELF TST/ERR indicator does not go out.

- Ensure that the NGIO board is properly installed.
- Replace the NGI board.
- Replace the NGIO board.

SELF TST/ERR indicator goes out, but the **PASS MOD TEST** (green) indicator does not light on one NGI board.

- Replace the NGI board, and then the board in the slot below it, if present.
- Replace the HPK2, K2LCN, or K4LCN board.

SELF TST/ERR indicator goes out, but the **PASS MOD TEST** (green) indicator does not light on any controller boards.

- Replace the HPK2, K2LCN, or K4LCN board.
- Remove the top NGI board, then apply power. Repeat for each board until the fault is cleared. Retry the extracted boards.

BUS TRAN ERR indicator is lit on the NGI board and the **DTAK TIME OUT** indicator is lit on the HPK2 board.

- Replace the NGI board and the board in the slot below it, if present, one at a time.
- Replace the HPK2, K2LCN, or K4LCN board.

BUS TRAN ERR indicator is lit on the NGI board, and the **BGAK TIME OUT** indicator is lit on the HPK2, K2LCN, or K4LCN board.

- Remove the top NGI board, then apply power. Repeat for each board until the fault is cleared. Retry the extracted boards.
- Replace the HPK2, K2LCN, or K4LCN board.

The **TX** (yellow) indicator indicates transmit activity. If the indicator is not on, troubleshoot the Plant Information Network (PIN) cables and other interfaces.

CAUTION

There is a test jumper at the front of the NGI board (near the indicator lamps that are described above). The jumper has two positions, **TEST** and **NORMAL**. The jumper should always be in the **NORMAL** position when the network is operating. Placing this jumper in the **TEST** position can disrupt operation of other nodes that are operating on the PIN.

3.3.11 PLCI Board Faults

NOTE

For additional information on troubleshooting the PLCI board, refer to the *PLC Gateway Planning, Installation, and Service* manual.

SELF TST/ERR indicator does not go out.

- Ensure that the PLCI I/O board is properly installed.
- Replace the PLCI board.
- Replace the PLCI I/O board.

SELF TST/ERR indicator goes out, but the **PASS MOD TEST** (green) indicator does not light on one controller board.

- Replace the PLCI board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

SELF TST/ERR indicator goes out, but comes back on after the HG personality has been loaded.

- Incorrect software release or incorrect configuration. Refer to the *Programmable Logic Controller Gateway Planning, Installation, and Service* manual.
- Replace the PLCI board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

BUS TRAN ERR indicator is lit on the PLCI board.

- Replace the PLCI board.
- Replace the HMPU, HPK2, K2LCN, or K4LCN board.

3.3.12 WSI2 Board Faults

NOTE

For additional information on troubleshooting the WSI2 board, refer to the *Application Module^X* manual.

Transaction Error (red LED) indicator is lit. This indicates a module bus communication error.

- Replace the HMPU, K2LCN, or K4LCN board.
- Replace the TPDG or TPDGX board.
- Replace the WSI or WSI2 board.

DISASSEMBLY/ASSEMBLY

Section 4

4.1 DISASSEMBLY

WARNING

To protect equipment from voltage transients and ensure reliable operation, always turn power off before removing or replacing boards or any other part on a Five/Ten-Slot Module.

Disassemble the module only to the extent necessary to service or replace defective parts. The following procedures may require the use of a blade-tip screwdriver, nut drivers, a personnel grounding strap, and antistatic fixtures and equipment. Item numbers called out in the procedures refer to ballooned numbers in the exploded module diagram, Figure 4-1, and are keyed to the item numbers in *Spare Parts*, Section 5.

1. Remove front panel (1) by turning two quarter-turn fasteners on the right side of the panel and swinging panel out.

CAUTION

All assemblies used in this module are likely to contain electrostatically sensitive devices. Use a personnel grounding strap and grounded work surfaces and equipment. Store and transport parts only in electrostatically safe containers.

2. Remove boards (2) by pulling out on retainers and sliding boards out of chassis.
3. Remove fan assembly (3) by loosening the two quarter-turn bail-headed fasteners and sliding the assembly out of the module.
4. Remove* fan (4) by removing four screws (7), washers (6) and nuts (5), or by drilling out the heads of the rivets. Disconnect fan connector from printed-wire board.
5. Remove power supply (8) by pulling retainers, and sliding out of chassis.
6. Remove* board (9) from power supply by removing nuts (10) and washers (11) and sliding the board out of connector.
7. Remove I/O boards (12) from rear chassis.

* The power supply circuit board and individual fans are not replaceable items on current production units; the complete power supply or fan assembly is furnished at the ORU level.

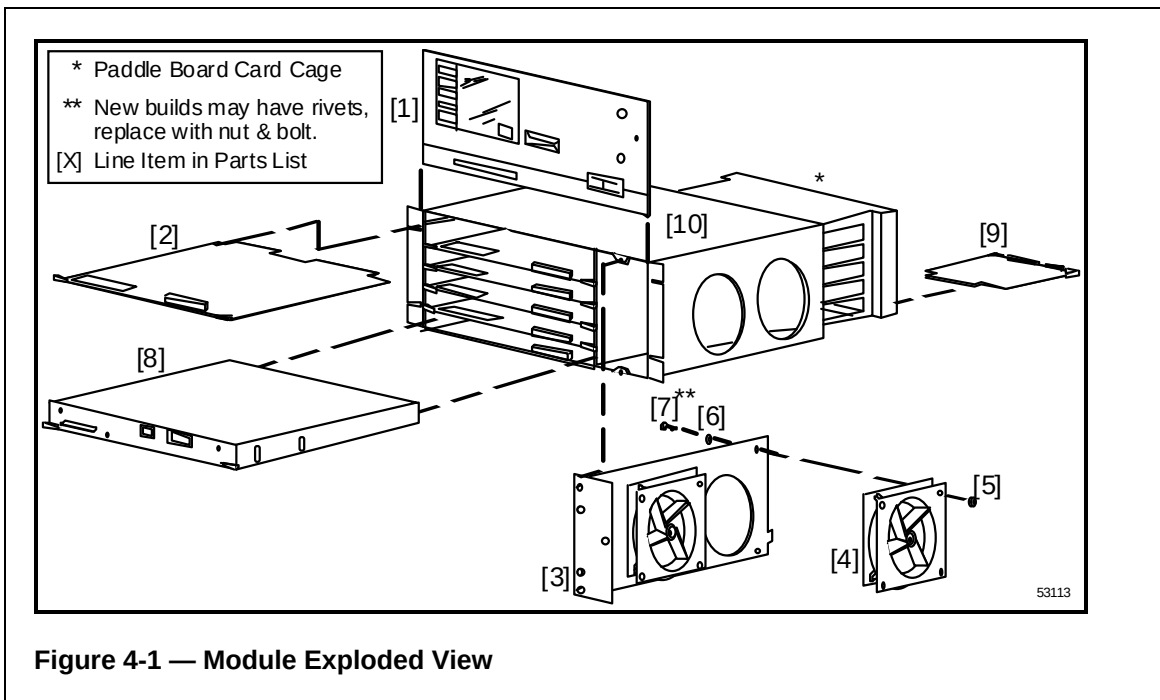


Figure 4-1 — Module Exploded View

NOTE

Before inserting a replacement board:

1. Make certain that the replacement board is at a revision level equal to or later than the board being replaced.
2. Make certain that any pins or switch settings (e.g., the node address on the LCN I/O board) match those on the board being replaced.

4.2 ASSEMBLY

1. Assemble I/O boards (9) to rear chassis, taking care not to damage connectors, and secure board retainers. Also, be sure that the I/O boards match the primary boards.
2. Insert power supply (8) into chassis and secure retainers.
3. Check air-flow direction arrow on fan and assemble* fan (4) to fan assembly and secure with screws (7), washers (6), and nuts (5). Connect fan connector to printed-wiring board.
4. Insert fan assembly (3) into chassis and secure bail-headed screws.
5. Insert boards (2) into chassis and secure retainers. If present, connect ribbon cables between VDG and PIC boards.
6. Assemble front panel (1) to chassis and secure quarter-turn fasteners.

* The power supply circuit board and individual fans are not replaceable items on current production units; the complete power supply or fan assembly is furnished at the ORU level.

SPARE PARTS

Section 5

5.1 INTRODUCTION

Table 5-1 lists spare parts for the Five-Slot and Ten-Slot Modules. Those parts at the Optimum Replaceable Unit (ORU) level are marked with an asterisk (*). The listing is arranged in item-number order with the items keyed to the call-out balloons in Figure 4-1. Also listed are those boards required for the various Five-Slot and Ten-Slot Module configurations, for more information refer to Tables 2-4 and 2-5. In some cases, more than one part is assigned to an item number (for instance any one of several board assemblies can occupy a slot in the module). In this case, the list is in alphabetical order.

5.2 SPARE PARTS LIST

Parts that are CE Compliant are identified in Table 5-1.

Table 5-1 — Parts List

Item	Description	Part Number
1	Five-Slot Module Cover Assembly	51400918-100
	Ten-Slot Module Cover Assembly	51400919-100
2	Refer to Table 2-4 in Section 2 for the specific board type	
3	5-Slot Fan Assembly (small)	* 51400646-100
	10-Slot Fan Assembly (large)	* 51400647-100
4	Fan, Tubaxial, DC	51190876-100
5	Nut, Hex, Lock, Nylon Insert, M 3.5	N/A
6	Washer, Flat, M 3.5	N/A
7	Mach Scr, Panhd, Cross Pt, M 3.5 X 20MM Long	N/A
8	Capacitor Alu Electro	51190064-416
	Capacitor Alu Electro	51190064-433
	Non-CE Compliant 8-Pin Power Supply (Five-Slot Module)	* 51196805-100
	Non-CE Compliant 8-Pin Power Supply (Ten-Slot Module)	* 51196810-100
	CE Compliant 3-Pin Power Supply (Five-Slot Module)	* 51196653-100
	CE Compliant 3-Pin Power Supply (Five-Slot Module)	* 51196654-100
	CE Compliant Power Supply Replacement Kits (includes 3-Pin power cords)	
	120/240 Vac (Five-Slot Module)	* 51402184-200
	120/240 Vac (Ten-Slot Module)	* 51402184-300

(Continued)

Table 5-1 — Parts List (Continued)

Item	Description	Part Number
9	Refer to Table 2-5 in Section 2 for the specific board type	
10	5-Slot Module for Classic Furniture	* 51195499-100
	5-Slot Module for Operator Interface Station (Ergonomic Furniture)	* 51197068-100
	5-Slot Module for Classic Furniture (CE Compliant)	* 51196692-100
	5-Slot Module for Ergonomic Furniture (CE Compliant)	* 51196729-100
	10-Slot Module	* 51108893-100
	10-Slot Module, 6/4 Split Node	* 51108893-200
	10-Slot Module, 6/4 Split Node (for Classic Furniture and Micro TDC 3000—CE Compliant)	* 51196730-100
MISC.	Mini Coax Cable Set (from PNM board to PNM I/O board)	51190896-100
	PIC/VDG Interface Cable	51303410-007
	PIC/VDG (also PNI/PNM) Interface Cable	51303627-003
	Non-CE Compliant 8-Pin Power Cord, 120 Vac	51303508-100
	Non-CE Compliant 8-Pin Power Cord, 240 Vac	51303508-200
	CE Compliant 3-Pin Power Cord, 120 Vac, Classic Furniture	51308093-300
	CE Compliant 3-Pin Power Cord, 240 Vac, Classic Furniture	51308094-300
	CE Compliant 3-Pin Power Cord, 120/240 Vac, Z-Furniture	51308092-200
	BNC Connector	51190728-105
	AM/HMPU Interface Cable (also for WSI/TPDG)	51304197-100
	AM Redundancy Cable	51304198-100

STARTUP Section 6

6.1 VISUAL CHECKS

- Check the module power and signal cables to ensure that they are present and tight.
- Check the ground strap at rear of chassis for a good connection.
- Ensure that all boards and the power supply are fully inserted into their connectors and that the board retainers are secured.

6.2 INITIALIZE MODULE

- Place the POWER switch in the on position. After 1.50 minutes or less, all red LED indicators should be off and all green LED indicators should be lit, and the alphanumeric display should display the LCN node address of the module. If not, check the display indicators and LED indicators in accordance with subsection 3.2, *Test Procedures*.
- If the module is not a History Module, press the **RESET*** button. Check the display indicators and LED indicators as above.

NOTE

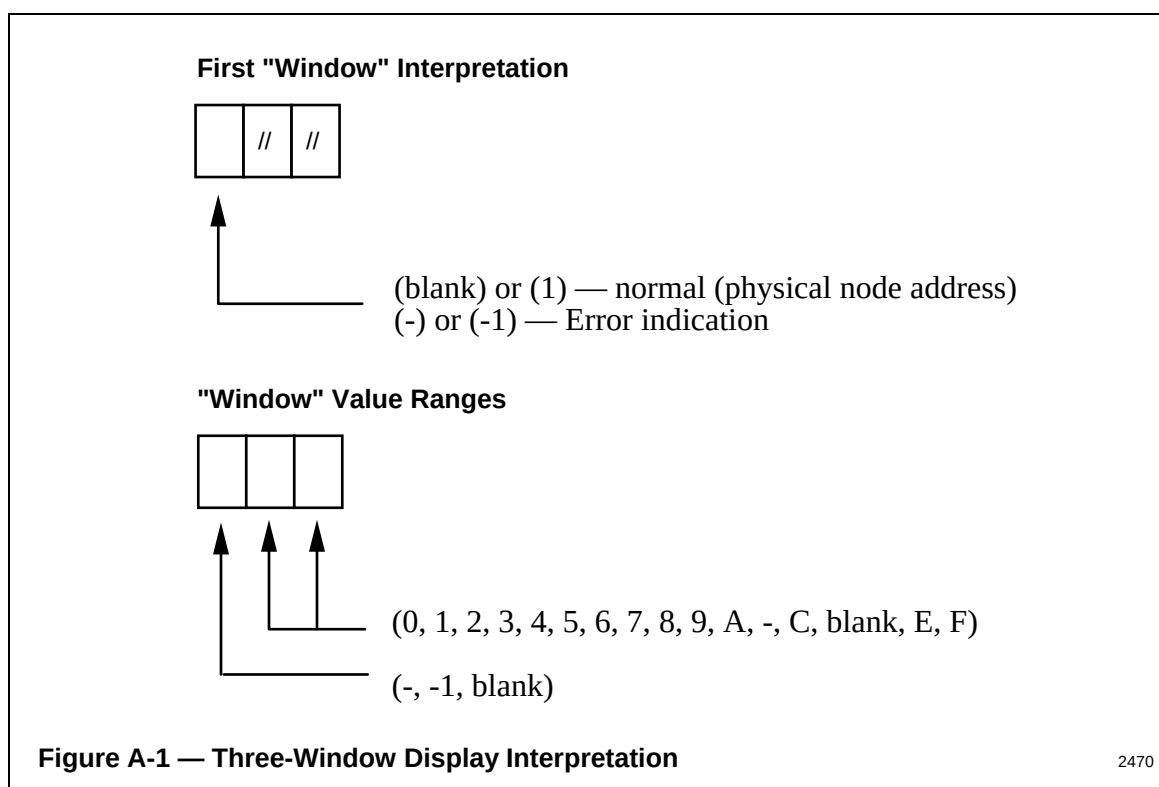
When you replace or restart a board or module with an outstanding maintenance recommendation, be sure to make an SMCC corrective action entry. Otherwise, the system will not start a recount of errors.

* **WARNING:** In the case of the History Module only, if you press the RESET button indiscriminately while the system is operating, you may cause damage to the data stored on the hard disk. See subsection 3.2 *Troubleshooting* in the *History Module Service* manual, part of the *Service* binder, for complete details.

ALPHANUMERIC DISPLAYS

Appendix A

The alphanumeric display enhances the man-machine interface by displaying hardware/firmware and software-driven messages. In interpreting the messages, special attention should be paid to the left-most element. As shown in Figure A-1, this element can have four distinct states, and the correct reading of the display is critical to understanding the message.



NOTE

The error code interpretations that follow do not apply while the node is executing test software such as HVTs.

A.1 RECOMMENDED ACTIONS FOR SPECIFIC CODE OCCURRENCES

Please note, when multiple recommendations are given for a window code-value, they are organized in order from most to least likely.

Recommendations marked with * indicate an unimplemented code-value. Any such occurrence should be brought to the attention of Honeywell TAC.

Window values marked as

//

 are "don't care" values.

-	0	//
---	---	----

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-	1	n
---	---	---

n=(-) Replace the HMPU/HPK2/K2LCN/K4LCN board; replace boards in slots 2 through 10 in turn.

n=(1 thru 7) Replace the HMPU/HPK2/K2LCN/K4LCN board.

n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-	2	n
---	---	---

n=(1 thru 7) Replace the board in slot 2; replace HMPU/HPK2/K2LCN/K4LCN board; replace the module backplane or chassis assembly.

n=(-) or (C) Replace the board in slot 2; replace HMPU/HPK2/K2LCN/K4LCN board; replace boards in slots 3 through 10 in sequence; replace the module backplane or chassis assembly.

n=(other) Replace the EMPU/HMPU/HPK2/K2LCN* board.

-	3	n
---	---	---

n=(1 thru 7),
(-) or (C) Replace the board in slot 3; replace the HMPU/HPK2/K2LCN/K4LCN board; replace the module backplane or chassis assembly.

n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-	4	n
---	---	---

n=(1 thru 7),
 (-) or (C) Replace the board in slot 4; replace the HMPU/HPK2/K2LCN/K4LCN board; replace the module backplane or chassis assembly.

n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN board.

-	5	n
---	---	---

n=(1 thru 7),
 (-) or (C) Replace the board in slot 5; replace the EMPU/HMPU/HPK2/K2LCN board; replace the module backplane or chassis assembly.

n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-	6	n
---	---	---

n=(1 thru 7),
 (-) or (C) Replace the board in slot 6; replace the HMPU/HPK2/K2LCN/K4LCN board; replace the module backplane or chassis assembly.

n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-	7	n
---	---	---

n=(1 thru 7),
 (-) or (C) Replace the board in slot 7; replace the HMPU/HPK2/K2LCN/K4LCN board; replace the module backplane or chassis assembly.

n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-	8	n
---	---	---

n=(1 thru 7),
 (-) or (C) Replace the board in slot 8; replace the HMPU/HPK2/K2LCN/K4LCN board; replace the module backplane or chassis assembly.

n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-	9	n
---	---	---

n=(1 thru 7), (-) or (C) Replace the board in slot 9; replace the HMPU/HPK2/K2LCN/K4LCN board; replace the module backplane or chassis assembly.

n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-	A	n
---	---	---

n=(A) or (C) Replace the EMPU/HMPU/HPK2/K2LCN board.

n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-	-	//
---	---	----

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-	C	//
---	---	----

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-		n
---	--	---

n=(A) If value persists beyond 30 seconds, replace the HMPU/HPK2/K2LCN/K4LCN board.

n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-	E	n
---	---	---

n=(0) An unexpected interrupt has occurred, replace the HMPU/HPK2/K2LCN/K4LCN* board.

n=(1) Replace the HMPU/HPK2/K2LCN/K4LCN board; replace boards in slots 2-10 in sequence.

n=(2) No working DRAM, replace or add memory board(s).

n=(3) Suspected program error, replace the HMPU/HPK2/K2LCN/K4LCN board.

n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-	F	//
---	---	----

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-1	0	n
----	---	---

n=(1 thru 7),
(-) or (C)

Replace the board in slot 10; replace the HMPU/HPK2/K2LCN/K4LCN board; replace the module backplane or chassis assembly.

n=(other)

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-1	1	//
----	---	----

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-1	2	//
----	---	----

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-1	3	//
----	---	----

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-1	4	//
----	---	----

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-1	5	//
----	---	----

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-1	6	//
----	---	----

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-1	7	n
----	---	---

- n=(0), (1) Check/replace the LCN cable.
- n=(2 thru 5) Replace the LCN interface board.
- n=(6) Replace the Winchester (drive or interface).
- n=(7 thru 9) Reload the node, replace the HMPU/HPK2/K2LCN/K4LCN board.
- n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN* board or another board in the node that could be hanging up the bus.

-1	8	n
----	---	---

- n=(0 thru 2) Replace the PIC board.
- n=(3) Reload the node, replace the HMPU/HPK2/K2LCN/K4LCN* board.
- n=(4,5,9) Reboot the system.
- n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-1	9	n
----	---	---

- n=(0,3,4,5) Reboot the system.
- n=(1 or 2) Record the error messages and call Honeywell TAC.
- n=(6) Use the SMCC to find defective board.
- n=(A) Replace the LCN/LLCN/K2LCN board/K4LCN board.
- n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-1	A	//
----	---	----

Replace the MCPU/EMPU/HMPU/HPK2/K2LCN/K4LCN* board.

-1	-	//
----	---	----

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-1	C	//
----	---	----

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-1		//
----	--	----

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-1	E	n
----	---	---

Replace the HMPU/HPK2/K2LCN/K4LCN* board.

n=(1 thru 9),
or (A) Replace board in corresponding slot number.

n=(other) Replace the HMPU/HPK2/K2LCN/K4LCN* board.

-1	F	n
----	---	---

n=(0 thru 8) Replace the HMPU/HPK2/K2LCN/K4LCN board.

n=(9) or (A) Replace the HMPU/HPK2/K2LCN/K4LCN* board.

n=(-) or (C) Replace the LCN interface board; if error repeats, check for a bus problem.

n=(E) Replace the memory boards; if multiple memory boards failed, check for a bus or interactive problems.

n=(F) Replace the HMPU/HPK2/K2LCN/K4LCN board.

n=(other) Check and/or replace the backplane or chassis assembly.

REPLACEMENT OF EMPU, HMPU, OR HPK2 BOARD WITH K2LCN

Appendix B

This appendix provides the procedure for replacing an EMPU, HMPU, or HPK2 processor board with a K2LCN board. It also provides the restrictions and conditions that apply for the replacement procedure.

B.1 INTRODUCTION

The procedure enables a trained technician to replace an EMPU, HMPU, or HPK2 processor board with a high-density, 68020-based K2LCN processor board in a Five-Slot or Ten-Slot Module LCN node. The K2LCN board requires software release 320 (or later).

To be a candidate for replacement, the module must not be a redundant Application Module (AM), which requires an HMPU board, or any processor board with special firmware, such as in a PCIM. The HMPU board has a coprocessor and the K2LCN board does not. If the module contains a Clock Source/Repeater (CS/R) board, the board cannot be located in the I/O slot behind the K2LCN board. It can be relocated to an unused I/O slot or another Five-Slot or Ten-Slot Module in the LCN coaxial segment.

If the module is one of a redundant pair, both processor boards must be replaced with K2LCN boards. The K2LCN board provides a processor, LCN interface, clock translation, and contains two, four, or eight megawords of on-board memory. It requires no external memory boards, so the procedure includes the removal of all memory boards from the module as well as the removal of the LCN board and the EMPU, HMPU, or HPK2 processor board.

WARNING

REPLACING AN EMPU, HMPU, OR HPK2 BOARD WITH A K2LCN BOARD REQUIRES REMOVING POWER FROM THE NODE AND MUST BE PERFORMED AT AN APPROPRIATE TIME TO AVOID DISTURBANCE TO THE PROCESS AND POTENTIALLY HAZARDOUS CONSEQUENCES TO EQUIPMENT AND PERSONNEL. DETERMINE IN ADVANCE THAT THE SCHEDULING OF THIS PROCEDURE IS IN ACCORDANCE WITH PROPER SYSTEM OPERATION.

WARNING

THE K2LCN BOARD CANNOT BE INSTALLED IN THE SAME MODULE SLOT AS A CLOCK SOURCE/REPEATER (CS/R) BOARD. INSTALLING A K2LCN BOARD IN THE SAME SLOT THAT A CS/R BOARD USES MAY DAMAGE THE BACKPANEL AND CAUSE THE ENTIRE MODULE TO FAIL.

B.2 RESTRICTIONS TO TOTAL REPLACEMENT

If you are contemplating the replacement of all the EMPU, HMPU, and HPK2 processor boards in Five-Slot or Ten-Slot Modules with K2LCN boards, be aware that the Clock Source/Repeater (CS/R) board can be the single-point ground connection for the LCN cable shield. Therefore, even if the 12.5 kHz (subchannel) clock function is not being used, a CS/R board or other grounding means is still required for each LCN coaxial segment.

A Dual Node Module can be included in the LCN coaxial segment to provide the single-point ground without requiring a CS/R board. Refer to the *Dual Node Module Service* manual for information. If there are no Dual Node Modules in the coax segment to provide a ground, then total replacement with K2LCN boards can only be accomplished by installing the CS/R board in an unused I/O board slot.

B.3 PREREQUISITES

The system must be up and running before performing the processor board replacement. You must have an LCN I/O board that is revision T, or later, (either in the module or as a spare board). If the board in the module is an LCNFL board, it (or an available spare board) must be a revision F or later.

B.4 NODE APPLICABILITY

Confirm that the K2LCN board replacement is applicable for the node:

1. Check to see if there is a Clock Source/Repeater (CS/R) board installed in Slot 1 at the rear of the module. If there is a CS/R board, the board must be moved to an unused I/O slot or another Five-Slot or Ten-Slot Module and change the Master and Slave Clock node numbers (System Wide Values, Clock Source) in the NCF. It cannot reside in the same slot as the K2LCN board.
2. Ensure that the amount of K2LCN board memory equals or exceeds the amount of memory on the processor board and/or the memory boards that the board is replacing.
3. The processor board in a redundant AM is an HMPU board. The HMPU board must not be replaced in a redundant AM by a K2LCN board.

If a nonredundant AM has an HMPU board and the External Load Module, AMCL02, is used for the coprocessor (on the third page of LCN Node Configuration), the HMPU board must not be replaced by a K2LCN board.

4. For performance compatibility, processor board types in redundant node pairs should not be mixed. When you must replace a processor board with a K2LCN board, and the affected node is one of a redundant pair, a K2LCN processor board must be installed in its partner's module also. See subsection B.6.

B.5 MEMORY SIZE

Determine the amount of process memory that the node has, including all memory boards and any memory on the processor board that is being replaced. The replacement K2LCN board must have at least the same amount of memory. Having more memory is not a problem. Because the K2LCN board is available with different memory sizes, there are three upgrade kits available.

K2LCN Board Upgrade Kit Description	Upgrade Kit Model Number	K2LCN Board Part Number
K2LCN Board – 2 Megawords of Memory	MP-ZPROC2	51402615-200
K2LCN Board – 4 Megawords of Memory	MP-ZPROC4	51402615-400
K2LCN Board – 8 Megawords of Memory	MP-ZPROC8	51402615-800

B.6 REDUNDANT MODULE REPLACEMENTS

This subsection applies only for a redundant Network Interface Module (NIM), Hiway Gateway (HG), or Enhanced Programmable Logic Controller Gateway (EPLCG). A redundant Application Module (AM) requires an HMPU processor board and is not eligible for replacement by a K2LCN board. Use the following procedure to prepare the module for disassembly.

1. Determine in which module, the primary or secondary node, that you want to perform the replacement first. The recommended procedure is to replace the processor board in the primary node first. This will require that you fail-over the primary to the secondary node, making the primary node available and allowing you to confirm proper operation of both nodes. It is acceptable to perform the replacement to the secondary node first, but you will not have validated the secondary node's proper operation prior to the processor board changes. In either case, you must be certain that the processor board is replaced in the node that is the backup node. When it is necessary to cause an intentional fail-over, follow the procedure in Step 2.
2. At a Universal Station or Universal Station^X, perform a shutdown of the node in which you will replace its processor board. This will cause an "orderly" shutdown to the "POWER ON" state and notify all the nodes that the node is being shut down, as well as cause the backup to take over as the active node. If the backup, now the active primary node, has successfully performed the "takeover," you are ready to proceed to Step 2 in subsection B.7. When subsections B.7 through B.10 are completed for the first module, repeat them for the second module.

B.7 DISASSEMBLY

Perform the following procedure to disassemble the module in which you are replacing the processor board.

1. If the node is “ON LINE,” go to a Universal Station and execute a shutdown of the node that will be disassembled. This will perform a “controlled” shutdown to the “POWER ON” state and notify all nodes that the node is being shut down.
2. Place the ac power switch on the front of the module in the off (0) position.
3. Remove the front cover by removing the screws; then swing the right side of the cover out 1 inch and pull it out to the right.
4. Be sure to observe proper ESD prevention practices, especially while handling boards during the following steps, including the reassembly procedure.
5. Remove the EMPU, HMPU, or HPK2, board and the LCN board from the front of the module. If present, remove all memory boards.
6. At the rear of the module, remove the LCN A and LCN B coax cables T-connectors from the LCN I/O board. Do **not** remove the cables from the T-connectors. Be sure that the system is still operational. Remove the LCN I/O board.

If the module is a single remote node that has an LCNFL board, the LCNFL board must be removed and checked to see if it is revision F (or later). If its revision is earlier than revision F, an LCNFL board at revision F (or later) must be substituted.

B.8 REASSEMBLY

Perform the following procedure to reassemble the module with a K2LCN board.

1. Check that the LCN I/O board you have is revision T (or later). Select the correct node address on the LCN I/O board and install it in slot 1 at the rear of the module. If a LCNFL board was replaced in Step 6 of the disassembly procedure because the board's revision was unacceptable, select the correct node address on the replacement LCNFL board, and install it in I/O Slot 1.

Reconnect the coax T-connectors, being careful to connect the A and B cables to their respective connectors on the board.

2. Be sure that there are no address jumpers on the LCN node address pinning block, TS2, at coordinate 9C on the earlier production K2LCN board, assembly 51401551-x00, or TS1 at coordinate 7A on the later production K2LCN board, assembly 51402615-x00. See Figures 2-10 and 2-11 for the location of the pinning blocks and the pinning instructions. Install the board in Slot 1 at the front of the module.
3. Configuration Rules require that no more than two empty slots exist between the K2LCN board and the next board. If there are more than two empty slots above the K2LCN board, you will now have to move the personality/peripheral board (EPDG, CLI, SIO, DHI, etc.) in Slot 5, and its associated I/O board at the rear of the module, down to Slot 4 or below. The preference is to install boards in the module from the bottom up, by installing the personality/peripheral board in Slot 2. An EPDG or TPDG board must have one or more empty slots above it. If your US or U^XS has Trend Pen Recorders, the SIO and SIO I/O boards should be placed between the K2LCN board and the EPDG or TPDG board.
4. Fasten the replacement kit label at the rear of the module chassis next to the original label.
5. Replace the module's front cover.
6. Place the ac power switch at the front of the module in the on (1) position.

B.9 TEST

Observe the indicators at the front of the module and especially on the K2LCN board. If a fault occurs, refer to Section 3 in this manual.

B.10 STARTUP

Restart the node by loading the node's specific personality and database as required. Refer to the *Engineer's Digest*.

REPLACEMENT OF EMPU, HMPU, HPK2, OR K2LCN BOARD WITH K4LCN Appendix C

This appendix provides the procedure for replacing an EMPU, HMPU, HPK2, or K2LCN processor board with a K4LCN board. It also provides the restrictions and conditions that apply for the replacement procedure.

C.1 INTRODUCTION

The procedure enables a trained technician to replace an EMPU, HMPU, HPK2, or K2LCN processor board with a high-density, 68040-based K4LCN processor board in a Five-Slot or Ten-Slot Module LCN node. The K4LCN board requires software release 500 or later.

To be a candidate for replacement, the module must not be a processor board with special firmware, such as in a PCIM. If the module contains a Clock Source/Repeater (CS/R) board, the board cannot be located in the I/O slot behind the K4LCN board. It must be relocated to an unused I/O slot or another Five-Slot or Ten-Slot Module in the LCN coaxial segment.

If the module is one of a redundant pair, both processor boards must be replaced with K4LCN boards. The K4LCN board provides a processor, LCN interface, clock translation, and contains four, eight, or sixteen megawords of on-board memory. It requires no external memory boards, so the procedure includes the removal of all memory boards from the module as well as the removal of the LCN board and the EMPU, HMPU, HPK2, or K2LCN processor board.

WARNING

REPLACING AN EMPU, HMPU, HPK2, OR K2LCN BOARD WITH A K4LCN BOARD REQUIRES REMOVING POWER FROM THE NODE AND MUST BE PERFORMED AT AN APPROPRIATE TIME TO AVOID DISTURBANCE TO THE PROCESS AND POTENTIALLY HAZARDOUS CONSEQUENCES TO EQUIPMENT AND PERSONNEL. DETERMINE IN ADVANCE THAT THE SCHEDULING OF THIS PROCEDURE IS IN ACCORDANCE WITH PROPER SYSTEM OPERATION.

WARNING

THE K4LCN BOARD CANNOT BE INSTALLED IN THE SAME MODULE SLOT AS A CLOCK SOURCE/REPEATER (CS/R) BOARD. INSTALLING A K4LCN BOARD IN THE SAME SLOT THAT A CS/R BOARD USES MAY DAMAGE THE BACKPANEL AND CAUSE THE ENTIRE MODULE TO FAIL.

C.2 RESTRICTIONS TO TOTAL REPLACEMENT

If you are contemplating the replacement of all the EMPU, HMPU, HPK2, and K2LCN processor boards in Five-Slot or Ten-Slot Modules with K4LCN boards, be aware that the Clock Source/Repeater (CS/R) board can be the single-point ground connection for the LCN cable shield. Therefore, even if the 12.5 kHz (subchannel) clock function is not being used, a CS/R board or other grounding means is still required for each LCN coaxial segment.

A Dual Node Module can be included in the LCN coaxial segment to provide the single-point ground without requiring a CS/R board. Refer to the *Dual Node Module Service* manual for information. If there are no Dual Node Modules in the coax segment to provide a ground, then total replacement with K4LCN boards can only be accomplished by installing the CS/R board in an unused I/O board slot.

C.3 PREREQUISITES

The system must be up and running before performing the processor board replacement. You must have an LCN I/O board that is revision T, or later, (either in the module or as a spare board). If the board in the module is an LCNFL board, it (or an available spare board) must be a revision F or later.

C.4 NODE APPLICABILITY

Confirm that the K4LCN board replacement is applicable for the node:

1. Check to see if there is a Clock Source/Repeater (CS/R) board installed in Slot 1 at the rear of the module. If there is a CS/R board, the board must be moved to an unused I/O slot or another Five-Slot or Ten-Slot Module and change the Master and Slave Clock node numbers (System Wide Values, Clock Source) in the NCF. It cannot reside in the same slot as the K4LCN board.
2. Ensure that the amount of K4LCN board memory equals or exceeds the amount of memory on the processor board and/or the memory boards that the board is replacing.
3. The processor board in a redundant AM is an HMPU board. An AMR board must be replaced by an EAMR board.

If a nonredundant AM has an HMPU board and the External Load Module, AMCL02, is used for the coprocessor (on the third page of LCN Node Configuration), the HMPU board can be replaced by a K4LCN board.

4. For performance compatibility, processor board types in redundant node pairs should not be mixed. When you must replace a processor board with a K4LCN board, and the affected node is one of a redundant pair, a K4LCN processor board must be installed in its partner's module also. See subsection C.6.

C.5 MEMORY SIZE

Determine the amount of process memory that the node has, including all memory boards and any memory on the processor board that is being replaced. The replacement K4LCN board must have at least the same amount of memory. Having more memory is not a problem. Because the K4LCN board is available with different memory sizes, there are three upgrade kits available.

K4LCN Board Upgrade Kit Description	Upgrade Kit Model Number	K4LCN Board Part Number	Memory Board Part Number
K4LCN Board – 4 Megawords of Memory	MP-ZPSC04	51402755-200	51201795-400
K4LCN Board – 8 Megawords of Memory	MP-ZPSC08	51402755-400	51201795-800
K4LCN Board – 16 Megawords of Memory	MP-ZPSC16	51402755-800	51201795-160

C.6 REDUNDANT MODULE REPLACEMENTS

This subsection applies only for a redundant Network Interface Module (NIM), Hiway Gateway (HG), Enhanced Programmable Logic Controller Gateway (EPLCG), and Application Module (AM). Use the following procedure to prepare the module for disassembly.

1. Determine in which module, the primary or secondary node that you want to perform the replacement first. The recommended procedure is to replace the processor board in the primary node first. This will require that you fail-over the primary to the secondary node, making the primary node available and allowing you to confirm proper operation of both nodes. It is acceptable to perform the replacement to the secondary node first, but you will not have validated the secondary node's proper operation prior to the processor board changes. In either case, you must be certain that the processor board is replaced in the node that is the backup node. When it is necessary to cause an intentional fail-over, follow the procedure in Step 2.
2. At a Universal Station or Universal Station^X, perform a shutdown of the node in which you will replace its processor board. This will cause an "orderly" shutdown to the "POWER ON" state and notify all the nodes that the node is being shut down, as well as cause the backup to take over as the active node. If the backup, now the active primary node, has successfully performed the "takeover," you are ready to proceed to Step 2 in subsection C.7. When subsections C.7 through C.10 are completed for the first module, repeat them for the second module.

C.7 DISASSEMBLY

Perform the following procedure to disassemble the module in which you will replace its processor board.

1. If the node is “ON LINE,” go to a Universal Station and execute a shutdown of the node that will be disassembled. This will perform a “controlled” shutdown to the “POWER ON” state and notify all nodes that the node is being shut down.
2. Place the ac power switch on the front of the module in the off (0) position.
3. Remove the front cover by removing the screws, then swing the right side of the cover out 1 inch, and pull it out to the right.
4. Be sure to observe proper ESD prevention practices, especially while handling boards during the following steps, including the reassembly procedure.
5. Remove the EMPU, HMPU, HPK2, or K2LCN board and the LCN board from the front of the module. If present, remove all memory boards.
6. At the rear of the module, remove the LCN A and LCN B coax cables T-connectors from the LCN I/O board. Do **not** remove the cables from the T-connectors. Be sure that the system is still operational. Remove the LCN I/O board.

If the module is a single remote node that has an LCNFL board, the LCNFL board must be removed and checked to see if it is revision F (or later). If its revision is earlier than revision F, an LCNFL board at revision F (or later) must be substituted.

C.8 REASSEMBLY

Perform the following procedure to reassemble the module with a K4LCN board.

1. Check that the LCN I/O board you have is revision T (or later). Select the correct node address on the LCN I/O board and install it in slot 1 at the rear of the module. If an LCNFL board was replaced in Step 6 of the disassembly procedure because the board's revision was unacceptable, select the correct node address on the replacement LCNFL board and install it in I/O Slot 1.

Reconnect the coax T-connectors, being careful to connect the A and B cables to their respective connectors on the board.

2. Be sure that there are no address jumpers on the LCN node address pinning block, TS2, at coordinate 21E on the earlier production K4LCN board, assembly 51401946-100, or TS1 at coordinate 7A on the later production K4LCN board, assembly 51402755-100. See Figures 2-12 and 2-13 for the location of the pinning blocks and the pinning instructions. Install the board in Slot 1 at the front of the module.
3. Configuration Rules require that no more than two empty slots exist between the K4LCN board and the next board. If there are more than two empty slots above the K4LCN board, you will now have to move the personality/peripheral board (EPDG, CLI, SIO, DHI, etc.) in Slot 5, and its associated I/O board at the rear of the module, down to Slot 4 or below. The preference is to install boards in the module from the bottom up, by installing the personality/peripheral board in Slot 2. An EPDG or TPDG board must have one or more empty slots above it. If your US or U^XS has Trend Pen Recorders, the SIO and SIO I/O boards should be placed between the K4LCN board and the EPDG or TPDG board.
4. Fasten the replacement kit label at the rear of the module chassis next to the original label.
5. Replace the module's front cover.
6. Place the ac power switch at the front of the module in the on (1) position.

C.9 TEST

Observe the indicators at the front of the module and especially on the K4LCN board. If a fault occurs, refer to Section 3 in this manual.

C.10 STARTUP

Restart the node by loading the node's specific personality and database as required. Refer to the *Engineer's Digest* manual.

UCLM: CCP AND UCIM REDESIGNED Appendix D

CCP (51402426) and UCIM (51402436) boards have been redesigned and are now known as UCLM-51306409. UCLM is a combination of CCP and UCIM.

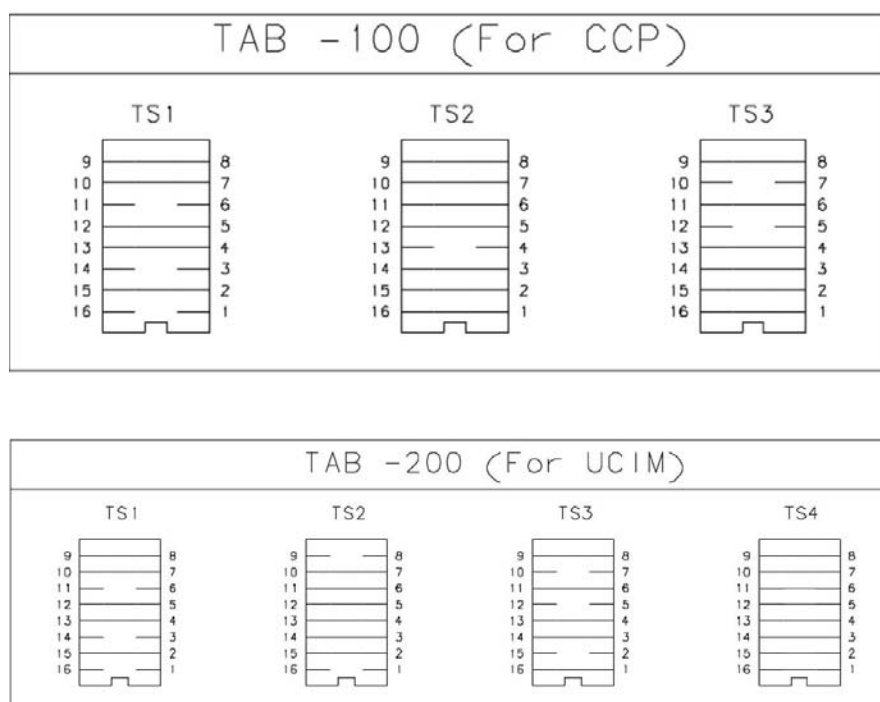
D.1 ORDERING INFORMATION

To order the boards use the Honeywell part numbers as provided in the following table:

Part Number	Board Name
51306409-100	CCP
51306409-200	UCIM

D.2 SWITCH SETTINGS OF UCLM BOARD

Figure C-1 illustrates the redesigned CCP and UCIM settings.



There is no change in the procedure to load AM personality on UCLM. Procedure remains the same as it was in CCP and UCIM.

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